MA411 Notes 1

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1 Recap: Discrete Probability

The purpose of this section is partly to remind you of a few basic concepts in probability that we will need throughout the course, and mainly to introduce some notation and motivation for the next section on measure theory.

Definition 1 (Probability space: objects)**.** *A* probability space $(\Omega, \mathcal{F}, \mathbb{P})$ *consists of three things.*

There is a sample space Ω*. This is 'all the things that can happen'.*

There is a event space $\mathcal{F}$*. This is a collection of subsets of* Ω*; it is 'all the events we know about'.*

There is a probability function $\mathbb{P} : \mathcal{F} \to [0, 1]$*. This says how likely events are.*

In a first course in probability, often $\mathcal{F}$ isn't mentioned, or it's briefly stated that we take $\mathcal{F}$ to be the power set $\mathcal{P}(\Omega)$ (i.e. all subsets of Ω). Most of our examples here are like this, but we also give a couple of important examples where $\mathcal{F}$ is *not* $\mathcal{P}(\Omega)$.

Example 2. *A fair six-sided die is rolled to generate a number.*

Here $\Omega = \{1, 2, 3, 4, 5, 6\}$, and $\mathcal{F} = \mathcal{P}(\Omega)$. The 'fair' bit tells us that $\mathbb{P}(X) = \frac{1}{6}|X|$ for all $X \in \mathcal{F}$. If your favourite event $X \subseteq \Omega$ contains 3 numbers, then the probability the die will land on one of those three numbers is $\frac{3}{6}$. It may not seem interesting, but it's convenient to also keep in the 'trivial' events $\emptyset$ with probability 0 and Ω with probability 1.

Example 3. *A fair coin is tossed.*

We have $\Omega = \{H, T\}$ and $\mathcal{F} = \mathcal{P}(\Omega)$. The 'fair' bit again tells us that $\mathbb{P}(X) = \frac{1}{2}|X|$ for all $X \in \mathcal{F}$.

Example 4. *A fair coin is tossed five times.*

We presume that the outcome of one coin toss does not affect the others: they are *independent*. We have $\Omega = \{H, T\}^5$ (the Cartesian product), and $\mathcal{F} = \mathcal{P}(\Omega)$, and $\mathbb{P}(X) = \frac{1}{32}|X|$.

Example 5. *What happens if an unfair coin is tossed once, or five times?*

Suppose the coin has probability $p \in [0,1]$ of landing Heads.

For one toss, the sample space is still $\{H,T\}$ and the event space is still its power set. The only change, compared to Example 3, is that now $\mathbb{P}(H) = p$ and $\mathbb{P}(T) = 1-p$.

For five tosses, the sample space is (still) $\{H,T\}^5$ and the event space is its power set. The change from Example 4 is that now if $X = \{x\}$ is a singleton set, we have $\mathbb{P}(X) = p^s(1-p)^{5-s}$ where s is the number of Heads in the sequence x. We (still) can calculate the probability $\mathbb{P}(Y)$ for larger sets Y as

$$\mathbb{P}(Y) = \sum_{x \in Y} \mathbb{P}(\{x\})\,.$$

This is a *product probability space*; it's the product of five copies of the single-coin-toss probability space. We should stress what that means: the sample space is a Cartesian product of five copies of $\{H,T\}$, *and* the event space is a 'product' of five copies of the single-toss event space[1], and the probability function is a product of the single-toss function, in the sense that

$$\mathbb{P}\big(\{(H,T,T,T,H)\}\big) = \mathbb{P}\big(\{H\}\big) \cdot \mathbb{P}\big(\{T\}\big) \cdot \mathbb{P}\big(\{T\}\big) \cdot \mathbb{P}\big(\{T\}\big) \cdot \mathbb{P}\big(\{H\}\big)\,.$$

It is formally naughty to use the one letter $\mathbb{P}$ for the two different probability functions on the two different probability spaces. However, this *abuse of notation* is widespread and we will continue it. We will only use different letters for different probability functions when they share a sample and (maybe) event space.

So far, we have also been careful about having 'the correct bracketing'. But it is more normal to write the last equation as

$$\mathbb{P}(HTTTH) = \mathbb{P}(H) \cdot \mathbb{P}(T) \cdot \mathbb{P}(T) \cdot \mathbb{P}(T) \cdot \mathbb{P}(H)\,,$$

Here we aren't writing out the sequence formally, rather a *string* (which is formally quite acceptable, but don't do it with numbers or it will be confusing..!), and we drop the set brackets that should be present around the singleton sets (which is formally not OK). It's also normal to write a set description into the probability: $\mathbb{P}(\text{exactly 3 Heads}) = \binom{5}{3}p^3(1-p)^2$, without bothering to write set brackets (or a 'formal' set description).

We will also make use of this abuse of notation often: it is a lot more readable, but you as the reader need to keep in mind what things are elements, sets, or sets of sets.

Independence, conditional probability

Perhaps the most important concept in probability is *independence*. Two events X and Y are said to be *independent* if we have $\mathbb{P}(X \cap Y) = \mathbb{P}(X) \cdot \mathbb{P}(Y)$ (the probability both events occur is the product of the individual probabilities). More generally, a collection of events are independent if $\mathbb{P}\big(\bigcap_{i \in I} X_i\big) = \prod_{i \in I} \mathbb{P}(X_i)$ for all subcollections $(X_i)_{i \in I}$.

Independence will be used over and over to calculate probabilities. In a sequence of coin flips, the individual flips are independent. If two events depend on disjoint sets of flips, they will also be independent, and so on. Because independence is so important, whenever we have events X and Y we will want to think about $X \cap Y$.

[1] We'll return to this once we made clear what exactly we want of an event space. For now, the general definition is a bit too complicated.

Exercise 1. *Two fair coins are flipped. Consider the three events* $A = \{HT, HH\}$, $B = \{TH, HH\}$, $C = \{HH, TT\}$. *Prove that any pair of these events are independent, but the set of all three is not.*

It's important to remember this example (and that it generalises): knowing that all pairs, or all triples, of a collection of events are independent is not the same as knowing that the entire collection is independent.

Solution to Exercise 1:

Each of A, B and C has probability $\frac{1}{2}$. When we intersect any pair, we get $\{HH\}$, which has probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, so each pair is independent. But the intersection of all three events is still $\{HH\}$, which does not have probability $\frac{1}{8}$.

Definition 6 (Conditional probability: conditioning on an event)**.** *Suppose* X *and* Y *are events, and* $\mathbb{P}(Y) > 0$. *We define the* conditional probability $\mathbb{P}(X|Y) = \frac{\mathbb{P}(X \cap Y)}{\mathbb{P}(Y)}$.

Intuitively, this reflects partial information. If I tell you Y occurs, it changes your belief about how likely X is.

Warning: while this conditional probability is a number, later in this course we will condition on other things than events. In this case *the conditional probability will usually not be a number: it will be a random variable.*

Exercise 2. *Suppose that* $A_1, \ldots, A_n$ *are events, and* $\mathbb{P}(A_1 \cap \cdots \cap A_n) > 0$. *Is it true that they are independent if and only if the following holds: for each* $2 \leq k \leq n$, *we have* $\mathbb{P}(A_k | A_1 \cap A_2 \cap \cdots \cap A_{k-1}) = \mathbb{P}(A_k)$*?*

Although really we should write a set after the | in conditional probability, it's common to write a list of events when we mean they all hold (i.e. we condition on their intersection). So we would often write the last equation as $\mathbb{P}(A_k | A_1, A_2, \ldots, A_{k-1}) = \mathbb{P}(A_k)$.

Solution to Exercise 2:

This is false. The given condition is enough to show that the probability of any 'initial subset' $A_1, A_2, \ldots, A_k$ all being true is what you would expect from independence: but it doesn't deal with other subsets.

One counterexample is given by the uniform distribution on $\{1, \ldots, 6\}$ with $A_1 = \{1, 2, 3\}$, $A_2 = \{2, 3, 4, 5\}$, $A_3 = \{1, 3, 6\}$. Here we have $\mathbb{P}(A_2 | A_1) = \frac{2}{3} = \mathbb{P}(A_2)$, and $\mathbb{P}(A_3 | A_1, A_2) = \frac{1}{2} = \mathbb{P}(A_3)$, so the condition holds, but $\mathbb{P}(A_2 \cap A_3) = \frac{1}{6} \neq \frac{2}{3} \cdot \frac{1}{2}$.

Random variables, expectation

Suppose the sample space Ω is a finite set. A function $X : \Omega \to \mathbb{R}$ is called a *random variable*. It's worth noticing there is nothing 'random' about it; it is just a function.

The *expectation* of X is $\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\{\omega\})$. (Here we are implicitly assuming $\mathbb{P}(\{\omega\})$ makes sense, i.e. $\{\omega\}$ is in the event space. We'll revisit this later.)

Theorem 7 (Linearity of expectation)**.** *Suppose* X *and* Y *are random variables (on* Ω*), and* $\alpha, \beta \in \mathbb{R}$. *Then* $\mathbb{E}(\alpha X + \beta Y) = \alpha \mathbb{E}(X) + \beta \mathbb{E}(Y)$.

This is a standard fact, and it follows immediately by writing out the definition and rearranging (because Ω is a finite set).

Here is a nice application.

Theorem 8. *Given $n \in \mathbb{N}$, suppose $\mathcal{S}$ is a subset of $\mathcal{P}\big(\{1, 2, \ldots, n\}\big)$ which is an* antichain, *i.e. for any $S, S' \in \mathcal{S}$, if $S \subseteq S'$ then $S = S'$. Then we have*

$$\sum_{S \in \mathcal{S}} \binom{n}{|S|}^{-1} \leq 1 \,.$$

Proof. Given n and $\mathcal{S}$, pick a *uniform* random permutation π of $\{1, 2, \ldots, n\}$ (all $n!$ permutations are equally likely; the sample space Ω is the set of permutations.).

Look at the sets $C_0, \ldots, C_n$ defined as follows: $C_0 = \emptyset$, and for each $1 \leq i \leq n$ let $C_i = C_{i-1} \cup \{\pi(i)\}$. That is, we add the next element in our permutation to the previous set.

Let $X = \big|\big\{i : C_i \in \mathcal{S}\big\}\big|$. So X counts the number of sets in $C_0, \ldots, C_n$ which happen to land in $\mathcal{S}$.

On the one hand, we cannot ever have $X \geq 2$, because then would be some C_i and C_j which are both in $\mathcal{S}$, say with $i < j$. But then $C_i \subseteq C_j$ and $C_i \neq C_j$, contradicting that $\mathcal{S}$ is an antichain. So

$$\mathbb{E}(X) = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\{\omega\}) \leq \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) = 1 \,.$$

On the other hand, we can think of X differently. Given $S \in \mathcal{S}$, let $\mathbb{1}_S$ be the *Bernoulli* random variable which takes value 1 if $C_{|S|} = S$ and 0 otherwise. Counting the number of elements of X which happen to be in $C_0, \ldots, C_n$ we get the same answer X. This says

$$X = \sum_{S \in \mathcal{S}} \mathbb{1}_S \,, \quad \text{so} \quad \mathbb{E}(X) = \sum_{S \in \mathcal{S}} \mathbb{E}(\mathbb{1}_S) \,.$$

Now, we can calculate $\mathbb{E}(\mathbb{1}_S)$ easily: we just need to count the ways we can get $C_|S| = S$. This happens if and only if the first $|S|$ elements of π are S, in any of the $|S|!$ orders, and the last $n - |S|$ elements can then come in any of the $(n - |S|)!$ possible orders. This is $|S|!(n - |S|)!$ ways in total, each with a probability of $\frac{1}{n!}$.

$$\mathbb{E}(\mathbb{1}_S) = \frac{|S|!(n - |S|)!}{n!} = \binom{n}{|S|}^{-1} .$$

Plugging this in we get the claimed inequality. $\square$

An easy consequence: $\binom{n}{|S|}$ is maximised when S has size $\lfloor \frac{n}{2} \rfloor$. Therefore for any antichain $\mathcal{S}$ we have

$$1 \geq \sum_{S \in \mathcal{S}} \binom{n}{|S|}^{-1} \geq \sum_{S \in \mathcal{S}} \binom{n}{\lfloor n/2 \rfloor} = |\mathcal{S}| \binom{n}{\lfloor n/2 \rfloor} ,$$

and rearranging we have $|\mathcal{S}| \leq \binom{n}{\lfloor n/2 \rfloor}$. This is the best possible upper bound on the size of an antichain on $\{1, \ldots, n\}$, because there is an antichain of that size: all subsets of $\{1, \ldots, n\}$ of size $\lfloor n/2 \rfloor$.

The event space represents information

Example 9. *We generate a sequence of* 2 *symbols from* $\{H, T\}$ *as follows. We will first flip a fair coin to generate the first symbol. Then we will change to a second coin for the last symbol.*

The sample space is $\Omega = \{H, T\}^2$.

In this example, we are not told how the second coin behaves. It might be fair, it might not be; it might depend on the first coin flip; it might even be controlled by an Evil Adversary and not random at all.

We can define the event space $\mathcal{F}$ to capture what information we *do* have: what is going on with the first coin-flip. We write

$$\mathcal{F} = \Big\{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\Big\}.$$

Going with this, we have

$$\mathbb{P}(\emptyset) = 0\,, \quad \mathbb{P}(\{HH, HT\}) = \mathbb{P}(\{TH, TT\}) = \tfrac{1}{2}\,, \quad \mathbb{P}(\Omega) = 1\,,$$

defining the probability function.

The point of this example is the following: we don't have enough information to make sense of '$\mathbb{P}(\{HH\})$', for example; we don't know what the second coin does. Nor can we argue that $\mathbb{P}(\{HH, TT\})$ is $\frac{1}{2}$. That *would* be true if the second coin was a fair coin, or even if it's not a fair coin but comes up heads with some other probability p whatever the first coin does: but we don't know either of those is true. It could even be that the Evil Adversary always makes the second coin come up the same as the first coin did (then $\mathbb{P}(\{HH, TT\})$ would be 1). Because we don't know, we don't try to define the probability for these sets: the domain of $\mathbb{P}$ is $\mathcal{F}$.

Example 10. *We generate a sequence of* 10 *symbols from* $\{H, T\}$ *by flipping* 10 *fair coins in turn. We pause after we flipped the first* 5.

As before, the sample space is easy: it is $\{H, T\}^{10}$.

The obvious event space here is $\mathcal{P}(\Omega)$, which is a set of 1 024 sequences, each with the same probability.

However, let's look at another event space $\mathcal{F}$: all the events that are 'decided' when we pause. For example the event 'The third coin came up Heads' is decided; this is the set $\{\mathbf{x} \in \Omega \,:\, x_3 = H\}$. A non-example is 'The number of Heads is odd': we don't know if this event occurs or not after the fifth coin is flipped, we'll need to see all the flips. Another non-example is 'There was at least one Head': for most ways the first five coins come up, we already see a Head, but if we got $TTTTT$ then we still need to see the remaining coin flips before we know whether a Head comes up.

We'll return to this kind of example later: let's briefly say what $\mathcal{F}$ is. Let $\mathbf{y} \in \{H, T\}^5$. Then

$$\Big\{\mathbf{x} \in \{H, T\}^{10} \,:\, x_1 = y_1, x_2 = y_2, \ldots, x_5 = y_5\Big\}$$

is an element of $\mathcal{F}$, and so is the union of any number of such sets (and so is $\emptyset$), and that's it. It follows $\mathcal{F}$ contains $2^5 = 32$ sets.

This event space again represents information: it lets us separate the events we already know about after the first five coin flips from the events for which we need more information. We'll often want to think about some 'time passing'. At time 0, we haven't seen any coin flips. At time 5, we've seen the first 5 coin flips; $\mathcal{F}$ contains all the 'past' events at time 5, and everything else is a 'future' event. And $\mathcal{P}(\Omega)$ is what we know at time 10 (when the *process* ends) and all events are 'past'.

Discrete Probability Space

There are a couple of observations we can make:

(1) We want $\mathbb{P}(\Omega) = 1$, so $\Omega \in \mathcal{F}$.

(2) If $X \in \mathcal{F}$ is an event, either X or its *complement* $X^c = \Omega \setminus X$ must occur: so $\mathbb{P}(X) + \mathbb{P}(X^c) = 1$—which means we should have $X^c \in \mathcal{F}$. We say $\mathcal{F}$ is *closed under complements.*

(3) If $X, Y \in \mathcal{F}$ are *disjoint* events (i.e. $X \cap Y = \emptyset$) then we want $\mathbb{P}(X \cup Y) = \mathbb{P}(X) + \mathbb{P}(Y)$ (*finite additivity*), which means we should insist on $X \cup Y \in \mathcal{F}$. We say $\mathcal{F}$ is *closed under finite disjoint union.*

(4) If $X, Y \in \mathcal{F}$ are events, we want to discuss $\mathbb{P}(X \cap Y)$, the probability that X and Y both occur: so $X \cap Y \in \mathcal{F}$. We say $\mathcal{F}$ is *closed under finite intersection.*

The word 'finite' appears above because we have $X \cap Y \cap Z = (X \cap Y) \cap Z \in \mathcal{F}$ for events X, Y, Z, and so on (by induction) for any finite collection of events and also for disjoint unions and for additivity. What we want to stress is that we *cannot* handle infinite collections of events with these axioms.

Definition 11 (Discrete probability space: axioms)**.** *In a discrete probability space* $(\Omega, \mathcal{F}, \mathbb{P})$*, the event space* $\mathcal{F}$ *must contain* Ω *and be closed under complements, finite disjoint unions, and finite intersections. The probability function must satisfy* $\mathbb{P}(\Omega) = 1$ *and be finitely additive.*

The name for such a $\mathcal{F}$ is an *algebra* on Ω.

Exercise 3.

(a) Let $\mathcal{F}$ *be an algebra on* Ω*. Prove that for all* $A, B \in \mathcal{F}$ *we have* $A \cup B \in \mathcal{F}$ *and* $A \setminus B \in \mathcal{F}$*.*

(b) Suppose $\mathcal{F}$ *is a collection of subsets of* Ω *which is closed under complements and any (not necessarily disjoint) finite unions. Prove that it is closed under finite intersections.*

Solution to Exercise 3:

(a) We can write $A \setminus B = A \cap B^c$, where the latter uses the operations in the definition of 'algebra'. Similarly (having figured out that algebras are closed under set difference) we have $A \cup B = A \cup (B \setminus A)$.

(b) We can write $A \cap B = (A^c \cup B^c)^c$. By induction, we can take the intersection of any finite collection of sets in $\mathcal{F}$ and we obtain an element of $\mathcal{F}$.

It's standard to say an algebra is closed under complements and (any, not necessarily disjoint) finite unions. The previous exercise shows that this is equivalent to the definition we gave.

Philosophical comment

What *is* probability?

We'll take a 'formal' view: it's a number we can calculate according to certain formal rules, end of story.

There are (at least) two other positions here. 'Frequentism' says 'it's the proportion of times something happens in the long run'. Bayesianism says it's something to do with beliefs. If you are a philosopher, you can have fun knocking down any of these positions. The objection to the formal view is: how can you describe what your number means in the real world, not just saying it is a number which bears some relation to a bunch of different probability numbers?

It is worth considering what probability does and doesn't model well, though.

In practice, these formal numbers do seem to model well things like coin flips, dice rolls and various bits of physics—it doesn't make much sense to ask if these are 'really random', but at least the models give predictions that match the real world.

They are a bad model for lack of knowledge. People may often say 'it's a coin flip' in place of 'I have no idea', but: if you flip a fair coin a million times, I'll be happy to place a bet on your getting between four and six hundred thousand heads. I wouldn't want to place money on your guessing the truth of a million statements that you don't know much about, not even that you'll be right about half the time.

They are a dangerous model for poorly understood phenomena. Defaults on house mortgages, most of the time, seem to be pretty uncorrelated and modelling them as independent random events usually gives a good idea of what to expect. But if there is a severe shock to the housing market, these 'uncorrelated' events turn out to sometimes come all at once; the probabilistic model gives completely false predictions.

2 Recap: cardinality and real analysis

For the following section, we'll need some basic mathematical facts. If you are *not* happy with these (after a bit of study or remembering your undergraduate degree), now is the time to contact me and discuss whether this is the right course for you.

Cardinality

Given a set S, we say S is *countable* if there is a bijection between S and a subset of the natural numbers $\mathbb{N}$. (For us, a finite set is countable.) This is the same thing as saying that we can write $S = \{s_1, s_2, \dots\}$, where the dots may hide a finite or infinite number of terms: we can list the members of S.

Theorem 12. *If I is a countable set, and S_i is countable for each $i \in I$, then $\bigcup_{i \in I} S_i$ is countable.*

This is proved using the *diagonal enumeration* argument: draw a grid, listing the $i \in I$ going right along the top of the grid, and fill in each column of the grid by listing S_i downwards. Now draw diagonals in the direction that makes all of them finite (left and up) covering the grid. Finally, read off the elements of each diagonal we encounter going down in turn, skipping over any elements we previously read. This is a list of the union. It's worth noting that in this area, some textbooks will say that a statement is true 'because the countable union of countable sets is countable', when rather often there are no sets around and what the author means is 'because we can do diagonal enumeration'. I'll try to stick to 'by diagonal enumeration' when I mean that.

In particular many familiar sets, like the integers and the rationals, are countable.

Not every set is countable.

Theorem 13. *The real numbers $\mathbb{R}$ are uncountable.*

This is proved by the 'Cantor diagonal argument'[2] contradiction: suppose there is a listing of the reals, and write out a real number c whose ith decimal place differs from that of the ith real number, for each $i \geq 1$ in turn. This is easy to do, provided you are careful about the fact $0.999... = 1$ and similar things. Now the contradiction is this: c is a real number, so it must appear on the list, say in place n. But this cannot be, since c differs from the nth real in place n.

We say two sets A and B *have the same cardinality*, $|A| = |B|$, if there is a bijection between them. It's (fairly) easy to check that for example $[0, 1]$ has the same cardinality as $\mathbb{R}$; both are uncountable. We say $|A| \geq |B|$ if there is an injection from B to A. Justifying this notation, if $|A| \geq |B|$ and $|B| \geq |A|$ (there are injections in both directions) the Cantor-Schröder-Bernstein theorem says $|A| = |B|$ (there is a bijection). We write $|A| < |B|$ if $|A| \leq |B|$ and $|A| \neq |B|$.

Assuming the axiom of choice, cardinality is an order on sets (any two sets have the same cardinality, or one is larger than the other). But there are (infinitely) many different infinite cardinalities.

[2] Which is nothing to do with diagonal enumeration, even though both arguments are due to Cantor.

Theorem 14. *For every S, we have $|S| < |\mathcal{P}(S)|$.*

This is more or less the same Cantor diagonal argument. We clearly have an injection from S to $\mathcal{P}(S)$: map each $s \in S$ to $\{s\}$. So the possibility we need to exclude is $|S| = |\mathcal{P}(S)|$. Suppose this were true, and fix a bijection $f : S \to \mathcal{P}(S)$. We define a new set T by $T = \{s \in S : s \notin f(s)\}$. This is a member of $\mathcal{P}(S)$, so it is in the image of f: there is some $t \in S$ such that $f(t) = T$. But this is impossible, because it leads to a paradox: both $t \in T$ and $t \notin T$ contradict the definition of T, but one of the two must be true.

In particular, $\mathcal{P}(\mathbb{R})$ is bigger than $\mathbb{R}$.

Something important to notice is that cardinality is purely about sets and functions. It doesn't 'see' that $\mathbb{R}$ has an order, or a topology (the collection of open sets); it has nothing to do with analysis.

Real Analysis

We do not need much analysis.

Definition 15 (supremum, infimum)**.** *A real number u is an* upper bound *of $S \subseteq \mathbb{R}$ if $x \geq s$ for all $s \in S$. It is the* supremum *of S, $\sup(S)$, if it satisfies $u \leq u'$ for all upper bounds u' of S. Reversing the inequalities we get a* lower bound*, and the* infimum $\inf(S)$*, of S.*

The supremum is called a maximum $\max(S)$ *if $\sup(S) \in S$, and similarly we write the* minimum $\min(S)$ *in the case that $\inf(S)$ is in S.*

Recall the *least upper bound property* (LUBP) states that S has a supremum if and only if it is non-empty and has an upper bound; similarly the *greatest lower bound property* (GLBP) for infimum.

Definition 16 (limit of a sequence)**.** *A sequence $(a_1, a_2, \dots)$ of real numbers tends to a* limit *L (and this is unique) if we have*

$$\forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n > N : |a_n - L| < \varepsilon .$$

A *series*, written $\sum_{i=1}^{\infty} b_i$, means the limit (if it exists! if not it's called 'divergent') of the *partial sums* $a_n = \sum_{i=1}^{n} b_i$.

In this course, we usually care about convergent series whose terms are non-negative. These are well-behaved; more generally, any *absolutely convergent* series (i.e. the series obtained by replacing negative terms with their absolute values is convergent) is well-behaved.

Theorem 17. *Let $\pi : \mathbb{N} \to \mathbb{N}$ be a bijection, and suppose $\sum_{i=1}^{\infty} b_i$ is an absolutely convergent series. Then $\sum_{i=1}^{\infty} b_i = \sum_{i=1}^{\infty} b_{\pi(i)}$ exists.*

Furthermore, if $\sum_{i=1}^{\infty} a_{i,j} = b_j$ is a convergent series whose terms are non-negative for each j, and $L = \sum_{i=1}^{\infty} b_i$ is a convergent series, then the diagonal enumeration series

$$\sum_{i=1}^{\infty} \sum_{j=0}^{i-1} a_{i-j,j}$$

is convergent with limit L; if any of the input series are divergent, so is the diagonal enumeration series.

Finally, if $\sum_{i=1}^{n} c_i$ is any series of non-negative terms, then grouping the terms in any way (adding brackets) does not change whether the series is convergent or the limit: if any grouped series is convergent with a given limit, so is the original series and vice versa.

Proof. The first part is in every real analysis textbook, as is the final part. We'll do the second part, mainly to show a useful trick. Recall that if $\sum_{i=1}^{\infty} c_i$ is any convergent series of non-negative terms, then the partial sums are increasing and (by definition) tend to a limit, in particular they are bounded above by $\sum_{i=1}^{\infty} c_i$.

Given n, consider the partial sum $d_n = \sum_{i=1}^{n} \sum_{j=0}^{i-1} a_{i-j,j}$. This is a finite sum, so we can rearrange it as we like. In particular, we have

$$d_n = \sum_{i=1}^{n} \sum_{j=0}^{i-1} a_{i-j,j} \leq \sum_{i=1}^{n} \sum_{j=1}^{n} a_{i,j}$$

(the RHS is the first n terms of the first n series). This is because all terms are non-negative and all the terms on the LHS are in the RHS sum. Then we get

$$d_n = \sum_{i=1}^{n} \sum_{j=0}^{i-1} a_{i-j,j} \leq \sum_{i=1}^{n} \sum_{j=1}^{n} a_{i,j} \leq \sum_{i=1}^{n} \sum_{j=1}^{\infty} a_{i,j} \leq \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{i,j} = L\,,$$

where the two new inequalities both use that the partial sums are bounded by the limit, once for each of the first n series, and then for the series $\sum_{j=1}^{\infty} b_j$.

Now comes the useful trick. Given ε, we want to show there exists N such that for all $n > N$, we have $d_n \geq L - \varepsilon$ (this is what's left to prove in the definition of 'tends to a limit'). It's enough, because the d_n forms an increasing sequence, to find one N such that $d_N \geq L - \varepsilon$. We use the *summable errors trick*:

Let N_1 be such that $\sum_{i=1}^{N_1} b_i > L - \frac{1}{2}\varepsilon$, and let N_2 be such that for each $1 \leq j \leq N_1$ we have $\sum_{i=1}^{N_2} a_{i,j} > b_j - 2^{-j-1}\varepsilon$. These all come from the definition of the limit—formally we should write N_2 as the maximum of several terms from the N_1 different limits.

The point here is the 'errors' $\frac{1}{2}\varepsilon$ and $2^{-j-1}\varepsilon$ are 'summable'; if we add them up all up, we have a geometric sum which is at most ε.

Let $N = 2\max(N_1, N_2)$. Now d_N is a sum of each of the sums $\sum_{i=1}^{N_2} a_{i,j} > b_j - 2^{-j-1}\varepsilon$ for $1 \leq j \leq N_1$, and some more terms. Since all these terms are non-negative, we have

$$d_N \geq \sum_{j=1}^{N_1} \sum_{i=1}^{N_2} a_{i,j} > \sum_{j=1}^{N_1} \left(b_j - 2^{-j-1}\varepsilon\right) = \Big(\sum_{j=1}^{N_1} b_j\Big) - \sum_{j=1}^{N_1} 2^{-j-1}\varepsilon > L - \tfrac{1}{2}\varepsilon - \sum_{j=1}^{N_1} 2^{-j-1}\varepsilon > L - \varepsilon\,,$$

which is what we wanted to prove. □

Here we had a finite collection of errors, but (if it had been useful in the proof) even a series of errors $\sum_{i=1}^{\infty} 2^{-i}\varepsilon = \varepsilon$ is summable. Also, there is nothing special about using a geometric series; it's just a convenient way of choosing errors which we know how to sum.

We can also note another useful concept in the above proof: one can always think of a convergent series of non-negative numbers as being 'the main bit' (the first terms whose sum is close to the limit) and 'the leftover' which is an infinite series but whose sum is tiny. The reason to think this way is: we can rearrange the terms of the 'main bit' because it is simply a finite sum; while the 'leftover' is only a small error. This is sometimes called *Littlewood's philosophy*. We'll use this idea a lot in what follows: not only a series can be split into a finite 'main' term and a tiny (but infinite sum) error, the same holds for a countable disjoint union of sets.

Most of Theorem 17 is 'intuitively obvious', but it does need a proof; it's false without the conditions on the series. The reason why is this:

Theorem 18. *If $\sum_{i=1}^{\infty} c_n$ is convergent but not absolutely convergent, then for every $x \in \mathbb{R}$ there is a rearrangement of the terms of the series whose limit is x.*

We sketched the idea of this in lecture: if I want to rearrange to make the limit be 2025, I will take positive terms (in the order they appear in the series) till my partial sum exceeds 2025, then negative terms (again, in the series order) till it drops below, then positive terms again, and so on.

The following theorem allows us to compare non-negative series.

Theorem 19. *Suppose $\sum_{n=1}^{\infty} a_n$ is a convergent series of non-negative real numbers, and suppose $0 \leq b_n \leq a_n$ for all $n \in \mathbb{N}$. Then $\sum_{n=1}^{\infty} b_n$ converges to a limit in $\left[0, \sum_{n=1}^{\infty} a_n\right]$.*

Exercise 4. *Prove this theorem.*

Solution to Exercise 4:

For each $n \in \mathbb{N}$ we have

$$0 \leq \sum_{i=1}^{n} b_i \leq \sum_{i=1}^{n} a_i \leq \sum_{i=1}^{\infty} a_i$$

where the final inequality uses the (easy analysis result) that the partial sums of a non-negative series are bounded by the limit.

Thus the sequence whose nth term is $\sum_{i=1}^{n} b_i$ is monotone increasing (because all $b_i \geq 0$) and its terms are in $[0, \sum_{i=1}^{\infty} a_i]$, so (by an easy analysis result) so is its limit.

As usual with analysis, we should remember that ∞ is a useful symbol, but it is not a number and treating it as such will lead to error. What the above theorems, more or less, say is: we will be able to treat series of non-negative terms more or less like finite sums, although we will need to engage with the definition of a limit fairly often. That can go wrong as soon as we allow negative numbers in, so we will avoid that as much as we can.

We need one more, somewhat harder theorem (which talks about the usual distance on $\mathbb{R}^d$).

Theorem 20 (Heine-Borel Theorem)**.** *For any $d \in \mathbb{N}$, suppose S is a closed and bounded set in $\mathbb{R}^d$, and let $U_1, U_2, \dots$ be open sets whose union contains S. Then there is a finite J such that $\bigcup_{j \in J} U_j$ contains S.*

Exercise 5. *Prove this theorem. Hint: if not, then for each n there is $x_n \in S \setminus \bigcup_{i=1}^{n} U_i$. What can we say about the sequence $(x_1, x_2, \dots)$ if it tends to a limit? Recall the Bolzano-Weierstrass Theorem that any bounded sequence has a convergent subsequence.*

Solution to Exercise 5:

Following the hint, suppose there is no such finite set; in particular $U_1 \cup \cdots \cup U_n$ does not contain S, so let for each $n \in \mathbb{N}$ the point x_n be in $S \setminus \bigcup_{i=1}^{n} U_i$. Because S is bounded, by the Bolzano-Weierstrass Theorem it has a convergent subsequence $(x_{i_n})_{n\in\mathbb{N}}$ whose limit is y. Because S is closed and the convergent subsequence is in S, so $y \in S$.

Since $y \in S$, there exists $m \in \mathbb{N}$ such that $y \in U_m$, and since U_m is open, by definition there exists $\varepsilon > 0$ such that the ball of radius ε around y is contained in U_m. But on the other hand, because $(x_{i_n})_{n\in\mathbb{N}}$ tends to y, there exists N such that for all $n > N$ we have $\|x_{i_n} - y\| < \varepsilon$. Choose now $n = \max(N+1, m)$. Then $i_n \geq n \geq m$, so by definition x_{i_n} is not in U_m, but at the same time it is within ε of y. $\lightning$

The conclusion of the Heine-Borel Theorem is called *compactness*, and often written 'every open cover has a finite subcover'.

3 Measure theory

Probability spaces

When Ω is a finite set, the definitions of the previous section are all we need. But when Ω is not finite, we need more.

Example 21. *We toss a fair coin until it comes up Heads.*

The sample space is $\Omega = \{H, TH, TTH, TTTH, \dots\}$.

We can easily figure out what the probability of any one of these elements should be; all the singleton sets. The probability of $TTTH$ is $\frac{1}{16}$, for example. From the previous section, we'd expect this is enough: using the axioms we'll get the event space to be $\mathcal{P}(\Omega)$ and we can calculate the probability of any event.

This is false. Using the axioms from the previous section, we get $\mathcal{F}$ consisting of all subsets of Ω which are either finite, or have finite complement; and we can calculate the probability only of such sets. We can't, for example, get to the probability that we generate an even-length sequence (i.e. $\mathbb{P}(TH, TTTH, TTTTTH, \dots)$). To fix this, we need the following extra axiom.

Definition 22 (countable union, σ-algebra, countable additivity, generated σ-algebra)**.** *Suppose I is countable, and $A_i \in \mathcal{F}$ for each $i \in I$. Then $\bigcup_{i\in I} A_i$ is the* countable union*. This is just the normal union: we are stressing that I is countable. If $\bigcup_{i\in I} A_i$ is always in $\mathcal{F}$, we say $\mathcal{F}$ is closed under countable unions. If $\mathcal{F}$ is an algebra which is closed under countable unions, we say $\mathcal{F}$ is a* σ-algebra.[3]

If in addition whenever the A_i are pairwise disjoint, we have $\mathbb{P}\big(\bigcup_i A_i\big) = \sum_i \mathbb{P}(A_i)$, we say $\mathbb{P}$ satisfies countable additivity*.*

Finally, if $\mathcal{G}$ is any collection of subsets of Ω, we write $\sigma(\mathcal{G})$ for the smallest σ-algebra on Ω that contains $\mathcal{G}$; this is the σ-algebra generated by $\mathcal{G}$*.*

Exercise 6.

(a) Give an example of two σ-algebras $\mathcal{F}$ and $\mathcal{F}'$ on $\{1, 2, 3, 4\}$ whose union $\mathcal{F} \cup \mathcal{F}'$ is not a σ-algebra.

(b) Prove that for any sets Ω and I, if $\mathcal{F}_i$ is a σ-algebra on Ω for each $i \in I$, then $\bigcap_{i\in I} \mathcal{F}_i$ is a σ-algebra on Ω.

(c) Deduce that $\sigma(\mathcal{G})$ is well-defined.

Solution to Exercise 6:

(a) One example is $\{\emptyset, \{1, 2\}, \{3, 4\}, \{1, 2, 3, 4\}\}$ and $\{\emptyset, \{1, 3\}, \{2, 4\}, \{1, 2, 3, 4\}\}$. Both are easily seen to be σ-algebras, but if their union were a σ-algebra it would contain $\{1, 2\}$ and $\{1, 3\}$, so also the intersection $\{1\}$.

[3] The σ here is not a variable; it just indicates 'and we can do countable operations too'. It is really σ 'Sigma' for 'Summe' (German), which is 'Series'.

(b) It's convenient here to use the equivalent (much as in Exercise 3) definition that a σ-algebra is closed under (not necessarily disjoint) countable unions and complements, since then we have less conditions to check.

Given $F \in \bigcap_{i\in I} \mathcal{F}_i$, because each $\mathcal{F}_i$ is a σ-algebra so $F^c \in \mathcal{F}_i$, so by definition $F^c \in \bigcap_{i\in I} \mathcal{F}_i$.

Similarly, given $F_1, \dots \in \bigcap_{i\in I} \mathcal{F}_i$, each F_n is in each $\mathcal{F}_i$, so by definition $F_1 \cup \dots$ is in each $\mathcal{F}_i$ and so $\bigcap_{i\in I} \mathcal{F}_i$ as required.

(c) Suppose $\{\mathcal{F}_i : i \in I\}$ is the set of all σ-algebras on Ω containing $\mathcal{G}$. This set is non-empty because $\mathcal{P}(\Omega)$ is in it. Its intersection is by (b) a σ-algebra, and it contains $\mathcal{G}$. It is also contained in all σ-algebras on Ω containing $\mathcal{G}$, making it 'the smallest' (by set inclusion).

Definition 23 (Probability space: axioms)**.** *In a probability space* $(\Omega, \mathcal{F}, \mathbb{P})$ *we have that* $\mathcal{F}$ *is a* σ*-algebra, and* $\mathbb{P}$ *is countably additive and satisfies* $\mathbb{P}(\Omega) = 1$.

It's important to notice that even though the notation looks the same, $\sum_{i\in I} \mathbb{P}(X_i)$ above is very different to the finite sum $\sum_{x\in Y} \mathbb{P}(\{x\})$ from the previous section. The sum here is an infinite sum—a series—and it is by definition the limit of its partial sums. Proving things about it needs real analysis, not just arithmetic. One comment: by Theorem 17, we don't need to worry about 'what order do we put on I to take partial sums?'. Because I is countable, we can put an order on it and use that order: whatever order we choose we get the same answer for the limit.

A continuous space

At this point we've seen that we need closure under countable unions. Why not simply ask for closure under any unions, countable or not?

The set theory is fine ($\bigcup_{i\in I} X_i$ is well-defined even if I is uncountable), but what would have to go with it is 'uncountable' additivity $\sum_{i\in I} \mathbb{P}(X_i)$. We don't have a good way to make sense of this.

Example 24. *Pick a number in* $[0, 1)$ *uniformly at random.*

Here the sample space $[0, 1)$ is clear. It's *not* clear what the event space should be. But we can make some observations.

The probability of each $x \in [0, 1)$ should be the same—otherwise the distribution surely isn't 'uniform'. That probability can't be strictly positive: if $\mathbb{P}(\{x\}) = \varepsilon > 0$, then picking some more than ε^{-1} distinct points in $[0, 1)$ we get a set A with $\mathbb{P}(A) > 1$, which isn't allowed. So either $\mathbb{P}(\{x\})$ isn't defined—that is, singleton sets aren't in the sample space—or the probability is zero. We'll shortly see that the former isn't workable: so $\mathbb{P}(\{x\}) = 0$ for all $x \in [0, 1]$.

But now we have the following difficulty: for any event $Y \subseteq [0, 1]$, we can write $Y = \bigcup_{x\in Y} \{x\}$. This is a disjoint union, so we'd like to write for the probability function

$$\mathbb{P}(Y) = \sum_{x\in Y} \mathbb{P}(\{x\}) = \sum_{x\in Y} 0\,.$$

The right-hand side sum, however we'd like to interpret it, doesn't do what we want; the only way to interpret it that makes any sense is to say that this sum is zero (whatever Y is), but then we don't get the 'right' answers for $\mathbb{P}(Y)$. It doesn't make sense to talk about 'uncountable' additivity, and for that reason we also don't want to allow uncountable unions; we stick to countable additivity and countable unions.

That means that in order to talk about the uniform measure, we cannot just specify the probability of singleton sets and work from there. Instead, let's pick a family of sets we understand: half-open intervals $[a, b)$ with $0 \leq a < b \leq 1$. Intuitively, we'd like to have $\mathbb{P}\big((a, b)\big) = b - a$.

Definition 25 (Borel σ-algebra on $[0, 1)$)**.** *The* Borel σ-algebra on $[0, 1)$ *is* $\mathcal{B}([0, 1)) = \sigma(\{[a, b) : 0 \leq a < b \leq 1\})$.

In particular, if $x \in [0, 1)$ then $\{x\} \in \mathcal{B}([0, 1))$: take the countable intersection of sets $[x, x + 1/n) \cap [0, 1)$ for $n \in \mathbb{N}$.

We can define a probability space $([0, 1], \mathcal{B}([0, 1)), \mathbb{P})$ which satisfies $\mathbb{P}\big((a, b)\big) = b - a$ for all $0 \leq a < b \leq 1$—that is, we can define a probability function $\mathbb{P}$ with this property—but this will need some work. The problem is sorting out how to define $\mathbb{P}(X \cap Y)$ and prove that we define something which is countably additive. We will return to this later with the *Carathéodory Extension Theorem*.

For now, we suppose the required $\mathbb{P}$ exists and try to calculate the probability of some other sets.

Theorem 26. *If* $0 \leq a < b \leq 1$ *then* $\mathbb{P}([a, b]) = \mathbb{P}((a, b]) = \mathbb{P}((a, b)) = b - a$.

In the above, obviously if $b = 1$ the first two intervals aren't in $[0, 1)$ so don't have a probability: we should formally have excluded this case.

Proof. We already proved that $\mathbb{P}(\{a\}) = \mathbb{P}(\{b\}) = 0$ (formally, on the assumption the singleton sets are events, but then we proved they are in $\mathcal{B}([0, 1))$). We can write all four types of interval as the disjoint union of (a, b) with one or both of these singleton sets, so by finite additivity it follows all four types of interval have the same probability. □

With a bit more work, we can give an (infinite) procedure to calculate the probability of any open set $S \subseteq [0, 1)$. One way to do this is as follows: let $S_0 = S$. For each $n \geq 1$ in turn, divide $[0, 1)$ into 2^n disjoint open intervals of equal probability (so for $n = 1$ the intervals are $(0, \frac{1}{2})$ and $(\frac{1}{2}, 1)$) and the singleton set points at their left ends, for a total of 2^{n+1} pieces. Let $\mathcal{A}_n$ consist of all the pieces which are subsets of S_{n-1}, and let $S_n = S_{n-1} \setminus \bigcup \mathcal{A}_n$ (note $\mathcal{A}_n$ is a set of subsets of $[0, 1)$, so $\bigcup \mathcal{A}_n$ is its union, all the elements of $[0, 1)$ which are in any set of $\mathcal{A}_n$).

Exercise 7. *Prove that we can write any open set in* $[0, 1)$ *as the disjoint union of a countable collection of open intervals and singleton sets. Conclude that* $\mathbb{P}(S)$ *is the sum of a series with non-negative terms.*

Solution to Exercise 7:

We run the outlined procedure. Observe that each $\mathcal{A}_n$ contains at most 2^{n+1} pieces (either open sets or singleton sets), so the union $\mathcal{A} = \bigcup_{n\in\mathbb{N}} \mathcal{A}_n$ is a countable union of finite sets, so it is countable. By construction, its sets are disjoint: if $C \in \mathcal{A}$, then C is in some $\mathcal{A}_n$, and is therefore by definition disjoint from S_n: it therefore cannot intersect any $D \in \mathcal{A}_m$ with $m > n$. It also cannot intersect any $D \in \mathcal{A}_n$ with $D \neq C$ because the sets in $\mathcal{A}_n$ are some parts of a partition of $[0, 1)$.

It is clear by construction that $\bigcup \mathcal{A} \subseteq S$.

What remains to do is show that $\bigcup \mathcal{A} = S$. Suppose for a contradiction this fails: then there exists $y \in S$ which is not in $\bigcup \mathcal{A}$.

Now y is not of the form $m2^{-n}$ for any integer $m \geq 0$ and $n \in \mathbb{N}$, otherwise it would be (at latest) in $\mathcal{A}_n$. Because S is open, there is $\varepsilon > 0$ such that the ball of radius ε around y is contained in S. Choosing n such that $\varepsilon > 2^{-n}$, there exists an integer $m \geq 0$ such that $m2^{-n} < y < (m + 1)2^{-n}$ (here we use that y is not of the form $m2^{-n}$ to write the strict inequalities), and $\big(m2^{-n}, (m+1)2^{-n}\big)$ is contained in S. By construction, either this interval is chosen in $\mathcal{A}_n$, or the reason it is not is that there is some $m < n$ such that the interval intersects $\bigcup \mathcal{A}_m$. But in the latter case the entire interval is in $\bigcup \mathcal{A}_m$ by construction. Either way we get $y \in \bigcup \mathcal{A}$.

Finally, having written S as a disjoint countable union, we get by countable additivity $\mathbb{P}(S) = \sum_{A\in\mathcal{A}} \mathbb{P}(A)$ is a series with non-negative terms.

This procedure *doesn't* let us calculate the probability of $[0, 1) \setminus \mathbb{Q}$: that isn't an open set. But we can nevertheless do it.

Exercise 8. *Calculate* $\mathbb{P}([0, 1) \setminus \mathbb{Q})$.

Solution to Exercise 8:

This one is easy: we know the probability of any singleton set, and in particular all members of $[0, 1) \cap \mathbb{Q}$, is 0. Now $[0, 1) \cap \mathbb{Q}$ is their disjoint union, so by countable additivity (recalling $\mathbb{Q}$ is countable) we get $\mathbb{P}([0, 1) \cap \mathbb{Q}) = 0$. Finally $\mathbb{P}([0, 1) \cap \mathbb{Q}) + \mathbb{P}([0, 1) \setminus \mathbb{Q}) = \mathbb{P}([0, 1)) = 1$ by additivity, and so $\mathbb{P}([0, 1) \setminus \mathbb{Q}) = 1$.

It turns out $\mathcal{B}([0, 1))$ contains all the 'interesting' sets for this course.

Not all sets have a probability

One question we did not yet answer is: can't we just take our event space to be $\mathcal{P}([0, 1))$?

The answer is No (at least if we want our probability function to be uniform). The idea of the proof is this. We will split $[0, 1)$ into pieces: a countable collection of disjoint sets. We will argue that these pieces cannot have positive measure (much as we showed above that the singleton sets can't have positive measure) and then we will deduce (because there are only countably many pieces in the collection, whereas there are uncountably many singleton sets) that by countable additivity we have $\mathbb{P}([0, 1)) = 0$, which is a contradiction.

We haven't really yet said what a 'uniform' probability function should be. But one property it should have is *shift-invariance*. Suppose for some $t > 0$, we have any event $S \subseteq [0, 1 - t)$, so $\mathbb{P}(S)$ is defined. We say $\mathbb{P}$ has *shift-invariance* if (for all such choices of t and S) the set $S_t = \{s + t : s \in S\}$ is also an event and $\mathbb{P}(S_t) = \mathbb{P}(S)$.

Theorem 27 (Vitali's construction)**.** *There is no shift-invariant probability function on* $\mathcal{P}([0,1))$*.*

The idea is not too hard; what is a bit tricky is that the 'pieces' need a rather weird construction: the Vitali construction.

Proof. We define an equivalence relation on $[0,1)$ by $x \sim y$ if $x - y$ is a rational number. There are uncountably many equivalence classes, but (using the Axiom of Choice) we can pick one element from each class to form a set V.

Now for each $q \in \mathbb{Q}$, let $V_q = \{v + q \,:\, v \in V\} \cap [0,1)$. These are the countably many pieces (because $\mathbb{Q}$ is countable) in $[0,1)$. We need to check they are disjoint and cover $[0,1)$; this follows from the definition.

Suppose for a contradiction that $\mathbb{P}$ is a shift-invariant probability function on $\mathcal{P}([0,1))$: so in particular V, and all its subsets and shifts, are events.

Suppose some piece V_q has $\mathbb{P}(V_q) = \varepsilon > 0$. We can write $V_q = (V_q \cap [0, \frac{1}{2})) \cup (V_q \cap [\frac{1}{2}, 1)$, and one of these two sets must have probability at least $\frac{1}{2}\varepsilon$. Suppose it is the former $V'_q = V_q \cap [0, \frac{1}{2})$. For each rational number $0 < r < \frac{1}{2}$, the set $V'_{q+r} = \{v + q + r \,:\, v + q \in V'_q\}$ is a subset of $[0,1)$ contained in V_{q+r}, and by shift-invariance $\mathbb{P}(V'_{q+r}) = \mathbb{P}(V'_q) \geq \frac{1}{2}\varepsilon$. It follows that $\mathbb{P}(V_{q+r}) \geq \frac{1}{2}\varepsilon$ for each rational $0 < r < \frac{1}{2}$. In particular, choose more than $2\varepsilon^{-1}$ rational numbers r and take the disjoint union of the V_{q+r}. By additivity, we obtain a subset of $[0,1)$ whose measure is more than $2\varepsilon^{-1} \cdot \frac{1}{2}\varepsilon = 1$, but then $\mathbb{P}([0,1)) > 1$, which is a contradiction. (If the second half of V_q is the part with large probability, we do the same as above but shift back instead of forwards.)

At this stage we are left only with the possibility $\mathbb{P}(V_q) = 0$ for all $q \in \mathbb{Q}$. But this too is impossible: then we have, by countable additivity,

$$1 = \mathbb{P}([0,1)) = \mathbb{P}\Big(\bigcup_{q \in \mathbb{Q}} V_q\Big) = \sum_{q \in \mathbb{Q}} \mathbb{P}(V_q) = 0\,. \qquad \square$$

General measure spaces

We could equally well think of the probability of some $S \in \mathcal{B}([0,1))$ as being its length, or its *measure*. In this context, there is no particular reason to insist on the measure of the entire space being 1. But we will use different notation; we'll reserve the 'probability notation' for probability spaces when we are thinking about probability.

Despite the warning earlier, we will use the symbol ∞ a bit like a number which is larger than any real number. The length of $\mathbb{R}$, for example, should be ∞. We will even allow ourselves to do a bit of arithmetic with ∞: if we add any real number and ∞, we get ∞; if we multiply ∞ by any positive real number we get ∞. If we try to subtract ∞ from anything (even ∞) or multiply ∞ by negative numbers, something has gone badly wrong. We write $\mathbb{R}^{\infty}_{\geq 0}$ (the *extended non-negative reals*) for the set of non-negative real numbers together with the symbol ∞. We'll also slightly extend the definition of sup and inf: if S either has no upper bound or contains ∞, we say $\sup(S) = \infty$, while if $\inf(S \setminus \{\infty\})$ exists we say it is $\inf(S)$; if $S = \{\infty\}$ then $\inf(S) = \infty$.

Finally, if we have a series whose terms are non-negative and which diverges, we will say the limit is ∞. We expand the concept 'countably additive' to take this in to account.

Definition 28 (Measure space, measure)**.** *A* measure space *is* $(X, \mathcal{F}, \mu)$ *where* X *is a set,* $\mathcal{F}$ *is a* σ*-algebra on* X *whose elements are called* measurable sets*, and the* measure $\mu : \mathcal{F} \to \mathbb{R}^{\infty}_{\geq 0}$ *is a function which is countably additive and satisfies* $\mu(\emptyset) = 0$*.*

So a probability space is a measure space with $\mu(X) = 1$: the sample space is X, 'events' and 'measurable sets' are the same thing, and 'probability function' is 'measure'.

The condition $\mu(\emptyset) = 0$ is not quite redundant. We have $\emptyset \cap \emptyset = \emptyset$, so $\emptyset \cup \emptyset$ is a disjoint union and additivity implies $\mu(\emptyset) = 2\mu(\emptyset)$. So either $\mu(\emptyset) = 0$ (as we insist) or $\mu(\emptyset) = \infty$. In the latter case it follows by additivity that $\mu(S) = \infty$ for every $S \in \mathcal{F}$; this is a pathological space and we therefore avoid talking about it.

Most examples of measure spaces we will meet in this course that are not probability spaces start by taking X to be $\mathbb{R}$, with its standard topology (that means the open sets you met at undergraduate), and taking the events to be the Borel σ-algebra. We can write this in general:

Definition 29 (Borel σ-algebra)**.** *If* $(X, \mathcal{T})$ *is any topological space (i.e.* $\mathcal{T}$ *is the collection of 'open sets in* X*') then the* Borel σ-algebra *of* $(X, \mathcal{T})$ *is* $\mathcal{B}\big((X, \mathcal{T})\big) = \sigma(\mathcal{T})$*.*

Where the topology is 'standard', e.g. for $\mathbb{R}$, it's common to simply write $\mathcal{B}(\mathbb{R})$: keep in mind though that there are different topologies on $\mathbb{R}$ and they do not all generate the same σ-algebras. We should really write the topology explicitly. If you haven't seen 'topological space' before, don't worry: in this course, we'll only ever talk about (a subset of) $\mathbb{R}^d$ for some finite d, with its usual open sets. So you don't need to know the definition of 'topological space'.

In Exercise 7 we argued that our previous definition of $\mathcal{B}([0, 1))$ is equivalent to the definition above. The same proof shows that we can take the open intervals in $\mathbb{R}$ to generate the entire Borel σ-algebra, or the squares in $\mathbb{R}^2$ or boxes in $\mathbb{R}^d$, to generate their corresponding Borel σ-algebras. We mentioned above (but did not prove) that there exists a measure μ on $\mathcal{B}(\mathbb{R}^d)$ which assigns the 'right' measure to these generating sets: the *Borel measure on* $\mathbb{R}^d$. (For example $\mu((a, b)) = b - a$ in $\mathbb{R}$.)

Exercise 9.

(a) Show that $\mathbb{Q} \subseteq \mathbb{R}$ *is measurable and has Borel measure zero.*

Let $C_1 = [0, 1]$*. For each* $n \geq 2$*, let* C_n *be obtained from* C_{n-1} *by removing the middle third of each closed interval: that is, if* (x, y) *is a (maximal under inclusion) open interval in* C_{n-1}*, then we remove* $(\frac{2}{3}x + \frac{1}{3}y, \frac{1}{3}x + \frac{2}{3}y)$*.*

(b) Explain why C_n *is a union of disjoint closed intervals and has Borel measure* $(2/3)^{n-1}$*.*

Let $C = \bigcap_{n=1}^{\infty} C_n$*.*

(c) Prove that C *has Borel measure zero.*

(d) *Prove that C is uncountable. Hint: each interval in C_n is split into a 'left' and a 'right' interval in C_{n+1}: use this to construct a bijection with $\mathcal{P}(\mathbb{N})$. You will need the analysis fact that any 'nested' sequence $A_1 \supseteq A_2 \supseteq A_3 \dots$ of closed and bounded intervals in $\mathbb{R}$ has non-empty intersection.*

Solution to Exercise 9:

(a) We more or less did this in Exercise 8, using some calculations earlier. But let's run over the full logic, starting from the definition of Borel measure in terms of open sets. For any given $q \in \mathbb{Q}$, we have $(q-1, q+1) = (q-1, q) \cup \{q\} \cup (q, q+1)$ is a disjoint union. So $\{q\} = (q-1, q+1) \setminus \big((q-1, q) \cup (q, q+1)\big)$ is in $\mathcal{B}(\mathbb{R})$, and $2 = 1 + \mu(\{q\}) + 1$ so $\mu(\{q\}) = 0$. Now $\mathbb{Q}$ is the countable disjoint union $\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$, so $\mu(\mathbb{Q}) = 0$ by countable additivity.

(b) We start with a collection (of one) disjoint closed interval of measure 1, and at each step we replace each interval $[x, y]$ in our collection with $[x, \frac{2}{3}x + \frac{1}{3}y]$ and $[\frac{1}{3}x + \frac{2}{3}y, y]$. These are still closed intervals, and (because they are disjoint subsets of the original interval they are disjoint. The interval $[x, y]$ has measure $y - x$, and the two replacement intervals together have measure $\frac{2}{3}(y - x)$, so summing $\mu(A_n) = \frac{2}{3}\mu(A_{n-1})$. Now the two claims of (b) hold by induction.

(c) We have $C_1, C_2, \dots$ are (by construction) in $\mathcal{B}(\mathbb{R})$, so their countable intersection C is also in $\mathcal{B}(\mathbb{R})$ and so has a measure. Suppose for a contradiction $\mu(C) = \varepsilon > 0$, and choose n such that $(2/3)^{n-1} < \varepsilon$. Then $C \subseteq C_n$, so $\mu(C) + \mu(C_n \setminus C) = \mu(C_n) < \varepsilon$. But this is impossible, since $\mu(C) = \varepsilon$ now implies $\mu(C_n \setminus C) < 0$.

(d) One way to do this is to think of a 'tree': the 'root' is the single interval in C_1, whose children are the 'left' and 'right' intervals in C_2. Each of these has a 'left' and a 'right' child in C_3, and so on. Observe that each child interval is a closed interval contained in its parent.

Given now any $S \subseteq \mathbb{N}$, choose intervals as follows: let A_1 be the 'left' interval in C_2 if $1 \in S$, and otherwise the 'right' interval. Now for each $n \geq 2$, let A_n be the 'left' child of A_{n-1} if $n \in S$, and otherwise the 'right' child.

This gives us a nested sequence of closed and bounded intervals, whose intersection is therefore non-empty (by the analysis fact from the question) so contains some x_S; by construction x_S is in each A_n and therefore in each C_{n+1} and also in C_1, so it is in C.

Finally, if $S \neq S'$ then consider the smallest n which is in exactly one of S and S': then x_S is (w.l.o.g.) in the 'left' child of A_{n-1} and $x_{S'}$ is in the 'right' child: since those intervals are disjoint we have $x_S \neq x_{S'}$. We have now constructed a bijection between $\mathcal{P}(\mathbb{N})$ and a subset of C (in fact, with all of C, though we'd need more work to show this). Since $\mathcal{P}(\mathbb{N})$ is not countable by Theorem 14, neither is C.

The set C constructed in the above exercise is called a *Cantor set*.

Historically, this area of mathematics, measure theory, started with the quest for a general definition of 'length of a subset of $\mathbb{R}$': a shift-invariant measure. As we've seen, if we want a shift-invariant measure on $[0, 1)$ with total measure 1, then there will be subsets of $\mathbb{R}$ which are not measurable.[4] We've claimed (but not yet proved) that there is a shift-invariant measure on $\mathbb{R}$ which assigns measure $b - a$ to each open set (a, b): the Borel measure.

[4] Actually, our proof doesn't really need $\mu((a, b)) = b - a$. It just needs that there is a set whose measure is bigger than 0 but not ∞ to replace $[0, 1)$.

Only later Kolmogorov highjacked the tools of measure theory in order to do probability properly. Very unfortunately, he did not rename any terms. In particular, this means that we will later talk about things like 'the event A is measurable with respect to the σ-algebra $\mathcal{F}_t$', where t is a time parameter. It's natural to think that we are worried about whether A actually has a probability when we write something like this.

We should remember that in probability the rôle of a σ-algebra is to talk about *information*, so that this phrase means something like 'at time t, we have seen the outcomes of some random choices. There are still random choices to be made, but we have enough information to decide whether A has happened or not'. We will never need to worry about the possibility that a set we encounter does not have a probability.

Properties of measures

Theorem 30 (Countable linearity of measures)**.** *Let $(X, \mathcal{F}, \mu_i)$ be a measure space for each $i \in \mathbb{N}$. Suppose $\alpha_i > 0$ is a real number (i.e. not ∞) for each $i \in \mathbb{N}$. Then the function $\mu = \sum_{i=1}^{\infty} \alpha_i \mu_i$ is a measure on $\mathcal{F}$.*

Proof. Since obviously $\mu(\emptyset) = 0$, we just need to check countable additivity. Let $A_1, \dots$ be disjoint sets in $\mathcal{F}$ with union A. We want to prove

$$\sum_{i=1}^{\infty} \alpha_i \sum_{j \in \mathbb{N}} \mu_i(A_j) = \sum_{i=1}^{\infty} \alpha_i \mu_i\Big(\bigcup_{j \in \mathbb{N}} A_j\Big) = \mu\Big(\bigcup_{j \in \mathbb{N}} A_j\Big) = \sum_{j \in \mathbb{N}} \mu(A_j) = \sum_{j \in \mathbb{N}} \sum_{i=1}^{\infty} \alpha_i \mu_i(A_j)\,,$$

where the middle equality is the countable additivity we want, the terms outside this are filling in the definition of μ; and the first term uses that each μ_i is a measure so countable additivity applies. (The sum $\sum_{j \in \mathbb{N}}$ and $\sum_{i=1}^{\infty}$ are identical, I'm just using different notation to highlight the change in the order of summation.)

Because μ_i is a measure, all these series are series of non-negative terms.

We need to be somewhat careful to deal with infinities. If there exist i and j such that $\mu_i(A_j) = \infty$, then the RHS sum is obviously ∞, while for the left sum we have $\mu_i\big(\bigcup_{j \in \mathbb{N}} A_j\big) \geq \mu_i(A_j) = \infty$ so the LHS sum is also ∞.

This means all the terms are real numbers: but we still can't instantly apply real analysis theorems, because the inner sums might diverge. (Actually, Theorem 17 is written with enough room to allow for applying it at this point, but for some future applications we would need more care.)

If a RHS inner sum $\sum_{i=1}^{\infty} \alpha_i \mu_i(A_j)$ diverges, we can repeat essentially the same argument as above to say that the LHS outer sum is at least $\sum_{i=1}^{\infty} \alpha_i \mu_i(A_j)$ which we assumed diverges: so LHS and RHS are both ∞.

Similarly if an LHS inner sum $\sum_{j \in \mathbb{N}} \mu_i(A_j)$ diverges, then the RHS outer sum is term by term at least as great as $\sum_{j \in \mathbb{N}} \alpha_i \mu_i(A_j)$ which diverges, and again the LHS and RHS are both ∞.

Suppose then none of these happens: all the terms in all the sums are real numbers. Theorem 17 now applies, and says either both sums on the LHS and RHS diverge (so we

have countable additivity with both sides being ∞), or both converge and they converge to the same value (which proves countable additivity too). $\square$

Theorem 31 (Continuity of measure)**.** *Let $(X, \mathcal{F}, \mu)$ be a measure space. Suppose $A_1 \subseteq A_2 \subseteq \dots$ are in $\mathcal{F}$. Then we have*

$$\lim_{n\to\infty} \mu(A_n) = \mu\Big(\bigcup_{n\in\mathbb{N}} A_n\Big).$$

Similarly if $B_1 \supseteq B_2 \supseteq \dots$ are in $\mathcal{F}$ and $\mu(B_1) < \infty$, we have

$$\lim_{n\to\infty} \mu(B_n) = \mu\Big(\bigcap_{n\in\mathbb{N}} B_n\Big).$$

Proof. Since each A_n is in $\mathcal{F}$, the countable union is also, justifying that $\mu\big(\bigcup_{n\in\mathbb{N}} A_n \bigcap)$ exists.

Let $a_1 = \mu(A_1)$ and for each $n \geq 2$ let $a_n = \mu(A_n \setminus A_{n-1})$. Because the sets are nested, we have $\mu(A_n) = \sum_{i=1}^{n} a_i$ for each $n \in \mathbb{N}$ (finite additivity), which are the partial sums of $\sum_{i=1}^{\infty} a_i = \mu(\bigcup_{n\in\mathbb{N}} A_n)$ by countable additivity. By definition the limit of the partial sums is the limit of the series. Observe that if the series diverges (or a term is ∞) we also obtain $\mu(\bigcup_{n\in\mathbb{N}} A_n) = \infty$ because it is bounded below by each partial sum.

We can reduce the second part to the first, by letting $A_n = B_1 \setminus B_{n+1}$ for each $n \in \mathbb{N}$. We have $\bigcap_{n\in\mathbb{N}} B_n = B_1 \setminus \bigcup_{n\in\mathbb{N}} A_n$, and we can rearrange this to

$$\mu\Big(\bigcap_{n\in\mathbb{N}} B_n\Big) = \mu(B_1) - \lim_{n\to\infty} \mu(A_n) = \mu(B_1) - \lim_{n\to\infty} (\mu(B_1) - \mu(B_{n+1}) = \lim_{n\to\infty} \mu(B_{n+1}).$$

Here we use $\mu(B_1) < \infty$ to justify writing the first equality: it is not permitted to write $\infty - \infty$. (If you prefer, it's always permitted by additivity to write $\mu(C \cup D) = \mu(C) + \mu(D)$ for disjoint sets C and D, what is not permitted is to 'subtract ∞ from both sides' if it turns out $\mu(D) = \infty$ to get $\mu(C) = \mu(C \cup D) - \mu(D)$.) $\square$

Exercise 10. *Let $(X, \mathcal{F}, \mu)$ be a measure space, and suppose $A, B \in \mathcal{F}$. Prove that*

(a) If $A \subseteq B$ then $\mu(A) \leq \mu(B)$.

(b) If $\mu(A) < \infty$, then $\mu(X \setminus A) = \mu(X) - \mu(A)$.

(c) For any countable I and sets $A_i \in \mathcal{F}$ for $i \in I$ we have

$$\mu\Big(\bigcup_{i\in I} A_i\Big) \leq \sum_{i\in I} \mu(A_i).$$

Solution to Exercise 10:

(a) We have $\mu(B) = \mu(A) + \mu(B \setminus A)$, and $\mu(B \setminus A) \geq 0$ by definition of a measure.

(b) We can always write $\mu(X) = \mu(A) + \mu(X \setminus A)$, and we can rearrange this as desired provided $\mu(A) < \infty$.

(c) Without loss of generality suppose $I = \mathbb{N}$. Let $a_i = \mu(A_i)$, and let $b_i = \mu\big(A_i \setminus (A_1 \cup A_2 \cup \cdots \cup A_{i-1})\big)$ for each $i \in \mathbb{N}$. Then by Theorem 19 we have

$$\sum_{i\in I} \mu(A_i) = \sum_{i=1}^{\infty} a_i \geq \sum_{i=1}^{\infty} b_i = \mu\Big(\bigcup_{i=1}^{\infty} A_i \setminus (A_1 \cup A_2 \cup \cdots \cup A_{i-1})\Big) = \mu\Big(\bigcup_{i\in I} A_i\Big).$$

Here the inequality is Theorem 19, the next equality is countable additivity, and the final one is because the two sets are the same.

Recall that a measure space is a probability space if $\mu(X) = 1$, so for a probability space (b) says $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. Property (c) is often called the *union bound*, and we'll use it repeatedly. Often we will have a 'bad event' which we want to show is unlikely (maybe it is enough to show its probability is at most 0.1). There are many different, likely not disjoint, possible causes for this bad event—that is, there is a collection of sets whose union is the bad event—and we cannot easily calculate the exact probability of their union: but we can show each individual cause has tiny probability and apply the union bound to obtain the required upper bound.

Given *any* subset S of X, we can define the *inner measure* $\mu_*(S) = \sup_{\substack{A\in\mathcal{F}\\ A\subseteq S}} \mu(A)$, and similarly the *outer measure* $\mu^*(S) = \inf_{\substack{A\in\mathcal{F}\\ A\supseteq S}} \mu(A)$.

Exercise 11.

(a) Prove that existence of the inner and outer measures follows from the LUBP and GLBP respectively. Recall ∞ is not a number and so cannot be in a set to which the LUBP or GLBP is applied.

(b) Prove that if $S \in \mathcal{F}$ then $\mu_*(S) = \mu^*(S) = \mu(S)$, and that for all $S \subseteq X$ we have $\mu_*(S) \leq \mu^*(S)$.

(c) Prove that the sup and inf in the above definitions are in fact max and min, respectively. That is, there is $A \in \mathcal{F}$ with $A \subseteq S$ and $\mu_*(S) = \mu(A)$, and similarly for the min.

Hint: this proof needs engagement with the analysis definition. If your proof seems to identify a largest (by set inclusion) A such that $\mu_*(S) = \mu(A)$, it is wrong.

Solution to Exercise 11:

(a) For the inner measure, let $\mathcal{G} = \{A \in \mathcal{F} : A \subseteq S\}$. If there is $A \in \mathcal{G}$ with $\mu(A) = \infty$, then (as mentioned extending the definition of sup to allow ∞) by definition $\mu^*(S) = \infty$ exists. So suppose there is no such $A \in \mathcal{G}$. We look at the set $G = \{\mu(A) : A \in \mathcal{G}\}$. This is a non-empty set of real numbers, because $\emptyset \in \mathcal{G}$ and so $0 \in G$. If G has no upper bound, then (again by the extended definition of sup) we have $\mu(S) = \infty$ exists. So finally suppose G does have an upper bound: then G is a non-empty and bounded above set of real numbers. By the LUBP, $\sup(G) = \mu^*(S)$ exists.

The outer measure is analogous. We obtain a non-empty set $G = \{\mu(A) : A \in \mathcal{F}, S \subseteq A\}$ because $\mu(X)$ is in G. If there are no $A \in \mathcal{F}$ with $S \subseteq A$ and $\mu(A) < \infty$, this justifies $\mu_*(S) = \infty$ exists. Otherwise, taking the subset of real numbers in G we have a non-empty set bounded below by 0 which by the GLBP has an infimum.

(b) Suppose $S \in \mathcal{F}$, by definition $\mu(S)$ is a lower bound for $\mu_*(S)$. If in fact $\mu_*(S) > \mu(S)$, then we must have $\mu(S)$ is finite. Suppose for a contradiction we have $\mu_*(S) > \mu(S)$. Consider the number $\min\left(\frac{1}{2}(\mu(S) + \mu_*(S)), 1\right)$. This is a real number smaller than $\mu_*(S)$: to see this, we need to consider a couple of cases. If $\mu_*(S) < \infty$, then the first term is a real number strictly smaller than $\mu_*(S)$ and therefore the minimum is too. If $\mu_*(S) = \infty$ then the first term is ∞ and therefore the minimum evaluates to 1, which is indeed a real number strictly smaller than $\mu_*(S)$.

By definition of $\mu_*(S)$ as a supremum, there is some $A \in \mathcal{F}$ with $A \subseteq S$ and $\mu(A) \geq \min\left(\frac{1}{2}(\mu(S) + \mu_*(S)), 1\right) > \mu(S)$ (if this set didn't exist, then $\min\left(\frac{1}{2}(\mu(S) + \mu_*(S)), 1\right)$ would be an upper bound for the supremum). But this contradicts Exercise 10(a). A similar argument deals with $\mu^*(S)$.

Given now any $S \subseteq X$, suppose we have $\mu_*(S) > \mu^*(S)$. If $\mu_*(S)$ is finite, then we can choose real numbers a and b such that $\mu^*(S) < a < b < \mu_*(S)$, and by the respective definitions of inf and sup there is $A \in \mathcal{F}$ with $S \subseteq A$ and $\mu(A) \leq a$, and $B \in \mathcal{F}$ with $B \subseteq S$ and $\mu(B) \geq b$. But now we have $B \subseteq S \subseteq A$ yet $\mu(B) \geq b > a \geq \mu(A)$, again contradicting Exercise 10(a). If $\mu_*(S)$ and $\mu^*(S)$ are both ∞, there is nothing to prove; if $\mu_*(S) = \infty$ and $\mu^*(S) < \infty$, then we can repeat the above argument with $a = \mu^*(S) + 1$ and $b = \mu^*(S) + 2$.

(c) For $\mu^*(S)$, if $\mu_*(S) = \infty$ then X is the required set. If $\mu^*(S) = c < \infty$, then for each $n \in \mathbb{N}$ there is a set $A_n \in \mathcal{F}$ such that $S \subseteq A_n$ with $\mu(A_n) \leq c + \frac{1}{n}$ by definition of inf. Let $A = \bigcap_{n\in\mathbb{N}} A_n$. By Theorem 31 we have $\mu(A) = c$, and trivially $S \subseteq A$, as required.

For $\mu_*(S)$, if $\mu_*(S) = \infty$ then (using the extended definition of sup) either there is $A \in \mathcal{F}$ with $A \subseteq S$ and $\mu(A) = \infty$, which is the required set, or for each $n \in \mathbb{N}$ there is $A_n \in \mathcal{F}$ with $A_n \subseteq S$ and $\mu(A_n) > n$. In the latter case $A = \bigcup_{n\in\mathbb{N}} A_n$ is the required subset of S with measure ∞ by Theorem 31 (note $\mu(A_1) < \infty$ otherwise we would be in the former case). If on the other hand $\mu_*(S) = d$ is finite, by definition of sup we have a set $A_n \in \mathcal{F}$ such that $A_n \subseteq S$ with $\mu(A_n) \geq d - \frac{1}{n}$. As before, letting $A = \bigcup_{n\in\mathbb{N}} A_n$ we have by Theorem 31 $\mu(A) = d$.

It's worth explaining briefly why we cannot ask for a largest $A \subseteq S$. By (b), if S is measurable then we do get a largest set: $A = S$. But suppose we have a non-measurable set S: for example the Vitali construction V described a couple of sections ago. Whatever $A \subseteq S$ we take with $A \in \mathcal{F}$, because S is not measurable we have $A \neq S$. So there exists $x \in S \setminus A$. But (in this case—it's not true in all measure spaces, but it is for $\mathbb{R}$ with Borel measure) $\{x\}$ is measurable, so since $\mathcal{F}$ is an algebra we have $A \cup \{x\} \in \mathcal{F}$ and $A \cup \{x\} \subseteq S$. This tells us A cannot be the largest measurable subset of S: there doesn't exist such a thing.

This may seem impossible: why can't we just go on forever till we build up S? The answer is we can go on arbitrarily long; we can even take the countable union of all the sets we build, but still we can only add *countably many* points to A. If S has uncountably many more points than any of its measurable subsets (as it must, in this case!) then we still won't build up S.

For future reference, we should notice the 'reason why' (c) is true is that we are allowed to take countable unions. If we only knew $\mathcal{F}$ were an algebra, we wouldn't be able to duplicate this proof.

Our definition above of inner and outer measure here are standard. It's worth observing though that if $\mu(X) < \infty$ then an equivalent definition of inner measure is

$$\mu_*(S) = \mu(X) - \mu^*(S^c)\,.$$

Exercise 12. *Prove that this definition is equivalent.*

Solution to Exercise 12:

This really just uses the fact that if $A \in \mathcal{F}$ so is A^c, and that $\mu(X) = \mu(A) + \mu(A^c)$ by additivity. We can rearrange this equation to $\mu(A) = \mu(X) - \mu(A^c)$ if $\mu(X) < \infty$. Now $A \subseteq S$ if and only if $S^c \subseteq A^c$, so the inf in the definition of $\mu_*(S)$ goes over the complements of the sets in the sup of $\mu^*(S^c)$ and the measures are related as claimed.

Once we get to construction of measures (when we don't start with a measure but something weaker), we will see that this equivalent definition is the right one to generalise.

Lebesgue measure, measure completion

In this part and the next we take a brief detour into measure theory. This part's purpose is to allow us to be lazy in the future but retain rigour.

We can be somewhat more general than Borel measure. Given any measure space $(X, \mathcal{F}, \mu)$, suppose $A \in \mathcal{F}$ has $\mu(A) = 0$. We say A is a *μ-null set*. Given any null set, we can safely declare all its subsets to be null sets as well, and then take the generated σ-algebra $\bar{\mathcal{F}}$ (the *measure completion of $\mathcal{F}$ with respect to μ*). If $\mathcal{F}$ is a Borel σ-algebra $\mathcal{B}(X, \mathcal{T})$, the result is called the *Lebesgue σ-algebra* and we write $\mathcal{L}(X, \mathcal{T})$. In this course it will usually be $\mathcal{L}(\mathbb{R})$. (As with the Borel σ-algebra, we should really specify we mean $\mathbb{R}$ with its usual topology: but we won't bother.)

Formally, we say $N \subseteq X$ is μ-null if there exists $B \in \mathcal{F}$ with $\mu(B) = 0$ and $N \subseteq B$.

If $S \in \bar{\mathcal{F}}$ can be written as a disjoint union $S = A \cup N$ where $A \in \mathcal{F}$ and N is μ-null, we define $\bar{\mu}(S) = \mu(A)$. This $\bar{\mu}$ is called the *completion* of μ, and the resulting $(X, \bar{\mathcal{F}}, \bar{\mu})$ is a *complete measure space*.

Exercise 13. *Fill in the details: check that $\bar{\mu}$ is well-defined. (There may be many different ways to write $S = A \cup N$ as a disjoint union of a measurable set and a μ-null set.)*

Prove that if $(X, \mathcal{F}, \mu)$ is any measure space, the completion $(X, \bar{\mathcal{F}}\, \bar{\mu})$ *is a measure space.*

Prove that $\bar{\mu}$ is the unique measure on $\bar{\mathcal{F}}$ whose restriction $\bar{\mu}|_{\mathcal{F}}$ to $\mathcal{F}$ is μ.

Solution to Exercise 13:

To check well-definedness, suppose $S = A \cup N = A' \cup N'$ are disjoint unions where A and A' are in $\mathcal{F}$, and N and N' are μ-null. Then we would like to have $\bar{\mu}(S) = \mu(A) = \mu(A')$: we need to prove that necessarily $\mu(A) = \mu(A')$. Because N is null, there is $B \in \mathcal{F}$ with $N \subseteq B$ and $\mu(B) = 0$, and similarly $B' \in \mathcal{F}$ with $N' \subseteq B'$ and $\mu(B') = 0$. Then we have

$$A \setminus (A \cap B) \subseteq A' \cup N \subseteq A' \cup B$$

so by Exercise 10(a) and (c) we have $\mu(A) - \mu(A \cap B) \leq \mu(A' \cup N) \leq \mu(A' \cup B') \leq \mu(A') + \mu(B')$. Plugging in $\mu(B) = \mu(B') = 0$ (and so $\mu(A \cap B) = 0$ too) we get $\mu(A) \leq \mu(A')$. The

same argument, swapping prime and non-prime sets, gives $\mu(A') \le \mu(A)$, so $\mu(A) = \mu(A')$ as desired.

This simply proves the function $\bar{\mu}$ is well defined on $\bar{\mathcal{F}}$; it doesn't prove it is a measure. We still need to argue $\bar{\mathcal{F}}$ is a σ-algebra and $\bar{\mu}$ is countably additive (we obviously have $\bar{\mu}(\emptyset) = 0$). It's convenient to use the 'short' definition of σ-algebra: closed under complement and countable union.

Suppose $S_1, \dots$ are in $\bar{\mathcal{F}}$, and write $S_n = A_n \cup N_n$ with $N_n \subseteq B_n$ where $\mu(B_n) = 0$ for each $n \in \mathbb{N}$.

We have ${S_1}^c = X \setminus (A_1 \cup N_1) = X \setminus (A_1 \cup B_1) \cup (B_1 \setminus N_1)$. Now $X \setminus (A_1 \cup B_1)$ is in $\mathcal{F}$, and $B_1 \setminus N_1$ is a subset of B_1 and hence null: this verifies ${S_1}^c \in \bar{\mathcal{F}}$.

We also have $\bigcup_{n\in\mathbb{N}} S_n = \bigcup_{n\in\mathbb{N}} A_n \cup \bigcup_{n\in\mathbb{N}} N_n$, and $A = \bigcup_{n\in\mathbb{N}} A_n$ is in $\mathcal{F}$. Let $N = \bigcup_{n\in\mathbb{N}} S_n \setminus A$, and we have $N \subseteq \bigcup_{n\in\mathbb{N}} N_n \subseteq \bigcup_{n\in\mathbb{N}} B_n$. By Exercise 10(c) we have

$$\mu\Big(\bigcup_{n\in\mathbb{N}} B_n\Big) \le \sum_{n\in\mathbb{N}} \mu(B_n) = 0\,,$$

so N is null. This verifies $\bar{\mathcal{F}}$ is a σ-algebra. For countable additivity, suppose now the S_n are pairwise disjoint. Then also the A_n are pairwise disjoint, so we have

$$\bar{\mu}\Big(\bigcup_{n=1}^{\infty} S_n\Big) = \mu(A) = \sum_{n=1}^{\infty} \mu(A_n) = \sum_{n=1}^{\infty} \bar{\mu}(S_n)\,,$$

as required: here the middle equality uses countable additivity of μ.

Finally, uniqueness is easier. Suppose μ' is any measure on $\bar{\mathcal{F}}$ whose restriction to $\mathcal{F}$ is μ, and let $S \in \bar{\mathcal{F}}$. Write $S = A \cup N$ where $N \subseteq B$ with $\mu(B) = 0$, with $A, B \in \mathcal{F}$. By Exercise 10(a) we have

$$\mu(A) = \mu'(A) \le \mu'(A \cup N) \le \mu'(A \cup B) = \mu(A) + \mu(B \setminus A) = \mu(A) + 0 = \mu(A)\,,$$

so $\mu'(S) = \mu(A)$, verifying uniqueness.

When we do this, if $\mu(X) < \infty$ then the sets S in $\bar{\mathcal{F}}$ are all sets satisfying $\mu_*(S) = \mu^*(S)$.

Exercise 14. *Fill in the details: prove that if $(X, \mathcal{F}, \mu)$ is any measure space with $\mu(X) < \infty$, then*

$$\bar{\mathcal{F}} = \big\{S \,:\, \mu_*(S) = \mu^*(S)\big\}\,.$$

Explain what goes wrong with this if $\mu(X) = \infty$.

Solution to Exercise 14:

We need to prove both directions of the set inclusion. If $S \in \bar{\mathcal{F}}$ then $S = A \cup N$ where $N \subseteq B$ with $A, B \in \mathcal{F}$ and $\mu(B) = 0$. Then $A \subseteq S \subseteq A \cup B$, so

$$\mu(A) \le \mu_*(S) \le \mu^*(S) \le \mu(A \cup B) = \mu(A) + \mu(B \setminus A) = \mu(A) + 0 = \mu(A)$$

as required; here we used Exercise 11 (b) and Exercise 10 (a).

The other direction uses Exercise 11(c) to obtain sets A and B such that $A \subseteq S \subseteq B$ with $\mu_*(S) = \mu(A)$ and $\mu^*(S) = \mu(B)$. Then $\mu(B \setminus A) + \mu(A) = \mu(B)$ by additivity; since $\mu(A) \leq \mu(X) < \infty$ we can subtract $\mu(A)$ from both sides to obtain $\mu(B \setminus A) = 0$. Letting $N = S \setminus A$, we have $N \subseteq B \setminus A$ is a null set.

This fails if $\mu(X) = \infty$ in the second direction: and in fact there are counterexamples. It is easy to check that since the Borel measure on $\mathbb{R}$ is shift-invariant, so is the Lebesgue measure. So the Vitali construction V is not Lebesgue measurable (and indeed its inner and outer measures are different, though calculating the outer measure is hard). But $V \cup [10, \infty)$ is easily seen to have both inner and outer measure ∞. Nevertheless, $V \cup [10, \infty)$ is not in the Lebesgue σ-algebra, for if it were so would be $\big(V \cup [10, \infty)\big) \cap [0, 1) = V$.

If $\mathcal{F}$ is the Borel σ-algebra $\mathcal{B}(X)$ on $\mathbb{R}^n$ (or some subset of it) with μ the Borel measure, the completion is called the *Lebesgue measure*, and usually we write $\lambda : \mathcal{L}(X) \to \mathbb{R}^\infty_{\geq 0}$.

It turns out that the Lebesgue σ-algebra is generally 'much bigger' than the Borel σ-algebra. It can be proved that the cardinality of $\mathcal{B}(\mathbb{R})$ is the same as of $\mathbb{R}$, while because of the uncountable Cantor set C, which has $|\mathcal{P}(C)| = |\mathcal{P}(\mathbb{R})|$ subsets, the Lebesgue σ-algebra has cardinality $|\mathcal{P}(\mathbb{R})|$.

For our purposes, the 'point' of Lebesgue measure (or measure completion in general) is the following. Often, constructions in this area 'almost' work neatly, but we have to do something ad-hoc at a small set of points. See Exercise 7 for an example: we'd really like to just use open sets, but we needed some singleton sets too. It's usually easy to see that this small set of points is a Borel null set, and the 'something ad-hoc' will lead us to taking a subset which we'll need to be measurable[5]. If we stick to the Borel measure, that will at least need a proof and sometimes extra care in the construction for it to even be true. When we use the Lebesgue measure, it's automatically measurable.

Construction of measures

We now need to give the promised explanation of how to define the 'uniform probability function on $[0, 1]$', or more generally construct Lebesgue measure.

This comes in two parts: a 'general measure theory' part which proves we can *extend* something that behaves a bit like a measure to an actual measure, and a construction (for the specific case) of the 'something a bit like a measure'.

Pre-measures

The following definition formalises 'a bit like a measure'.

Definition 32 (pre-measure)**.** *Let $\mathcal{A}$ be an algebra on X, and suppose $\mu_0 : \mathcal{A} \to \mathbb{R}^\infty_{\geq 0}$ satisfies $\mu_0(\emptyset) = 0$ and the* countable pre-additivity *condition that for all $A_1, \dots \in \mathcal{A}$ pairwise disjoint sets such that $\bigcup_i A_i$ is in $\mathcal{A}$, we have $\mu_0(\bigcup_i A_i) = \sum_i \mu_0(A_i)$. Then we call μ_0 a* pre-measure.

The important point to note is we only ask $\mathcal{A}$ to be an algebra: it is *not* necessarily closed under countable unions but only finite unions. This is why we need to specify that

[5] In Exercise 7 the set in question is countable and so obviously measurable; this won't always be the case.

we are interested only in choices of A_i where the countable union turns out to be in $\mathcal{A}$. The advantage is that $\mathcal{A}$ can be a lot simpler than a σ-algebra, which makes it easier to construct a pre-measure.

Our aim is to add some sets to $\mathcal{A}$ to make it a σ-algebra $\mathcal{F}$, and find a measure μ on $\mathcal{F}$ which agrees with μ_0 on $\mathcal{A}$: this is 'extension'. So countable pre-additivity is an obvious necessary condition to be able to extend μ_0 to a measure.

Carathéodory Extension Theorem: probability measures

It's convenient to start with the special case $\mu_0(X) = 1$; we are trying to construct a probability measure.

Given any set S, a *countable covering* of S is $A_1, A_2, \dots \in \mathcal{A}$ such that $S \subseteq \bigcup_i A_i$.

Given any set $S \subseteq X$, we define the *outer pre-measure* by

$$\mu_0^*(S) = \inf_{\substack{A_1,\dots\in\mathcal{A} \text{ a countable} \\ \text{covering of } S}} \sum_{i=1}^{\infty} \mu_0(A_i)\,.$$

We refer to $\sum_{i=1}^{\infty} \mu_0(A_i)$ as the *size* of a countable covering. We define the *inner pre-measure* $\mu_{0*}(S) = 1 - \mu_0^*(S^c)$.

It's important to say that (in contrast to Exercise 11) the infimum here is really an infimum: in general it will *not* be a minimum. The running example we'll consider is $X = [0, 1)$ with the algebra (we'll prove that!) of finite unions of half-open intervals $[a, b)$, where we assign pre-measure $b - a$ to such an interval. This is the setup to construct the Lebesgue measure; and in Theorem 34 we'll see an example of a set with outer pre-measure 0 but which doesn't have any countable covering with size 0.

Our main theorem will be in general (i.e. not tied to this example) though.

We say a countable covering is *disjoint* if the sets in it are pairwise disjoint.

Exercise 15. *Suppose $A_1, \dots$ is a countable covering of S. Prove that there exists a countable covering $B_1, \dots$ of S which is disjoint, whose size is at most the size of $A_1, \dots$, and with $\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i$.*

Solution to Exercise 15:

We let $B_1 = A_1$ and for each $n \geq 2$ we let $B_n = A_n \setminus (A_1 \cup \dots \cup A_{n-1})$. By definition these sets are pairwise disjoint. Because $\mathcal{A}$ is an algebra, these sets are all in $\mathcal{A}$. It's clear that the union is the same: in one direction $B_n \subseteq A_n$ for each n, so $\bigcup_{n\in\mathbb{N}} B_n \subseteq \bigcup_{n\in\mathbb{N}} A_n$, on the other hand if $x \in \bigcup_{n\in\mathbb{N}} A_n$ then there is some smallest n such that $x \in A_n$, and by definition also $x \in B_n$. Finally, $0 \leq \mu_0(B_n) \leq \mu_0(A_n)$ for each $n \in \mathbb{N}$ by Exercise 10(a), and so by Theorem 19 the size of $B_1, \dots$ is at most that of $A_1, \dots$.

The idea now (as in the last section) is that if $\mu_{0*}(S) = \mu_0^*(S)$ then S 'should' be measurable with measure $\mu(S) = \mu_{0*}(S)$. To begin with, we justify that the inner pre-measure isn't larger than the outer pre-measure.

Lemma 33. *Given an algebra $\mathcal{A}$ on X and pre-measure μ_0 on $\mathcal{A}$ such that $\mu_0(X) = 1$, for any $S \subseteq X$, we have $\mu_{0*}(S) \leq \mu_0^*(S)$.*

Proof. Suppose for a contradiction $\mu_{0*}(S) = \mu_0^*(S) + \varepsilon$, where $\varepsilon > 0$. By definition, there is a countable covering $A_1, \dots$ of S with size at most $\mu_0^*(S) + \frac{1}{4}\varepsilon$, and another $B_1, \dots$ of S^c with size at most $1 - \mu_{0*}(S) + \frac{1}{4}\varepsilon$. Thus $A_1, B_1, A_2, B_2, \dots$ is a countable covering of X with size at most $1 - \mu_{0*}(S) + \frac{1}{4}\varepsilon + \mu_0^*(S) + \frac{1}{4}\varepsilon = 1 - \frac{1}{2}\varepsilon$. It follows there exists a disjoint countable covering $C_1, \dots$ of X with size at most $1 - \frac{1}{2}\varepsilon$. But now countable pre-additivity of μ_0 states $\sum_{i=1}^{\infty} \mu_0(C_i) = \mu_0(X) = 1$, a contradiction. $\square$

Before we go on, it's worth getting a bit of intuition for countable covers. Think about the uniform measure on $[0, 1)$. We will certainly want $\mu([0, 1) \cap \mathbb{Q})$ to end up being 0 from this procedure: so there should exist countable covers of this set of small size. Intuitively, this seems ridiculous: $\mathbb{Q}$ is dense in $[0, 1)$, how can we possibly cover every rational point in (for example) an open set and not have the total size of these open sets be at least 1?

Theorem 34. *For each $\varepsilon > 0$, the set $\mathbb{Q} \cap [0, 1)$ has a countable cover with half-open intervals of size at most ε.*

The idea here is: for each $\frac{a}{b} \in \mathbb{Q} \cap [0, 1)$, choose a half-open interval centred on $\frac{a}{b}$ with length $f(b)$ decreasing as b increases. Any such construction clearly covers $\mathbb{Q} \cap [0, 1)$: choosing lengths appropriately, we can make the sum of the lengths as small as we like.

Exercise 16. *Fill in the details: what is $n(b)$ the number of rational points in $[0, 1)$ with denominator b? and given $\varepsilon > 0$, what should we choose for $f(b)$ to get $\sum_{b=1}^{\infty} f(b)n(b) < \varepsilon$?*

Solution to Exercise 16:

This is the proof of Theorem 34.

We have $n(b) = b$; the points are $\frac{0}{b}, \dots, \frac{b-1}{b}$. So we need to choose $f(b)$ such that $\sum_{b=1}^{\infty} bf(b) < \varepsilon$. For example, take $f(b) = 2^{-b-1}b^{-1}\varepsilon$, and then the sum becomes

$$\sum_{b=1}^{\infty} bf(b) = \sum_{b=1}^{\infty} 2^{-b-1}\varepsilon = \tfrac{1}{2}\varepsilon\,.$$

Our countable covering of $\mathbb{Q} \cap [0, 1)$ is then: for each $b \in \mathbb{N}$ and $0 \le a \le b - 1$ integer, let $A_{a,b}$ be the half-open interval

$$\left[\tfrac{a}{b} - 2^{-b-2}b^{-1}\varepsilon, \tfrac{a}{b} + 2^{-b-2}b^{-1}\varepsilon)\right]$$

and our countable covering is the collection of all $A_{a,b}$. By construction this indeed covers $\mathbb{Q} \cap [0, 1)$ (the rational $\frac{a}{b}$ is in $A_{a,b}$), and the calculation above shows its size is $\frac{\varepsilon}{2}$.

This proves that $\mu_0^*\big(\mathbb{Q} \cap [0, 1)\big) = 0$, by definition of the infimum. But there is no countable covering of $\mathbb{Q} \cap [0, 1)$ with size 0: to have a countable covering, there must be a non-empty set in our covering (obviously!) and this set by definition contains some non-empty half-open interval $[a, b)$, so $b > a$. But already this one interval has pre-measure $b - a > 0$, so the szie of the covering is at least $b - a > 0$. This justifies that the infimum in the definition of $\mu_)^*$ is (sometimes!) really an infimum and not a minimum.

Before we move on to the main theorem, one more definition. We say a countable covering $A, A_1, \dots$ of a set S is *ε-good* if $\sum_{i=1}^{\infty} \mu_0(A_i) \le \varepsilon$.

Exercise 17. *Prove that if $B_1, \dots$ is a countable covering of S with finite size (i.e. $\sum_{i=1}^{\infty} \mu_0(B_i)$ converges) then there exists an ε-good countable covering of S with at most the same size and the same union. Hint: the required idea is in the proof of Theorem 17.*

Solution to Exercise 17:

Suppose $B_1, \dots$ is a countable covering of S with $\sum_{i=1}^{\infty} \mu_0(B_i) = s$ convergent, and let $\varepsilon > 0$ be given.

By definition, there exists $n \in \mathbb{N}$ such that the partial sum $\sum_{i=1}^{n} \mu_0(B_i) > s - \varepsilon$. Since the terms of the series are non-negative, this implies $\sum_{i=n+1}^{\infty} \mu_0(B_i) < \varepsilon$.

We let $A = B_1 \cup \dots \cup B_n$ and for each $i \geq 1$ we let $A_i = B_{n+i}$. It's obvious this new countable covering has the same union, and indeed $A \in \mathcal{A}$ since $\mathcal{A}$ is an algebra. By Exercise 10(a) we have $\mu_0(A) \leq \mu_0(B_1) + \dots + \mu_0(B_n)$. Therefore the ith partial sum in the size, $\mu_0(A) + \mu_0(A_1) + \dots + \mu_0(A_{i-1})$, is upper bounded by $\sum_{i=1}^{n+i-1} \mu_0(B_i)$, which in turn is upper bounded by the size of $B_1, \dots$, and it follows the size of $A, A_1, \dots$ is at most that of $B_1, \dots$ as required. Finally we have $\sum_{i=1}^{\infty} \mu_0(A_i) = \sum_{i=n+1}^{\infty} \mu_0(B_i) < \varepsilon$ since the sums are identical by definition. Thus $A, A_1, \dots$ is ε-good.

If $A, A_1, \dots$ is an ε-good countable covering, we call A the *main set* and $A_1, \dots$ the *leftover*.

Formally, the assertion 'the same size' here doesn't require rearranging the terms of a series: however in what follows a lot of similar assertions will involve rearranging terms. Since we are dealing with convergent series whose terms are non-negative, this is permitted by Theorem 17.

We now justify the comment about 'tiny error'.

Lemma 35. *Given an algebra $\mathcal{A}$ on X and pre-measure μ_0 on $\mathcal{A}$ such that $\mu_0(X) = 1$, let $S \subseteq X$ and suppose $\mu_{0*}(S) = \mu_0^*(S) = p$. Suppose that $A, A_1, \dots$ is an ε-good countable covering of S with size at most $p + \varepsilon$, and $B, B_1, \dots$ is an ε-good countable covering of S^c with size at most $1 - p + \varepsilon$.*

Then $\mu_0(A), 1 - \mu_0(B) \in [p - \varepsilon, p + \varepsilon]$ and $\mu_0(A \cap B) \leq 4\varepsilon$.

Proof. The size of $A, A_1, \dots$ is $\mu_0(A) + \sum_{i=1}^{\infty} \mu_0(A_i)$. This number is at most $p + \varepsilon$ by assumption, and at least p because $\mu_0^*(S) = p$. Since $0 \leq \sum_{i=1}^{\infty} \mu_0(A_i) \leq \varepsilon$, we have $\mu_0(A) \in [p - \varepsilon, p + \varepsilon]$ as required. Similarly $\mu_0(B) \in [1 - p - \varepsilon, 1 - p + \varepsilon]$, as required.

Observe $A \setminus B, B, A_1, B_1, A_2, B_2, \dots$ is a countable covering of X whose size is $\mu_0(A \setminus B) + \mu_0(B) + \sum_{i=1}^{\infty} \mu_0(A) + \mu_0(B) \geq \mu_0(X) = 1$. The series here converges to a number between 0 and 2ε by Theorem 17, so $\mu_0(A \setminus B) + \mu_0(B) \geq 1 - 2\varepsilon$, and we conclude $\mu_0(A \setminus B) \geq p - 3\varepsilon$. Since $A = A \setminus B \cup (A \cap B)$ is a disjoint union, by additivity we have $\mu_0(A \cap B) = \mu_0(A) - \mu_0(A \setminus B) \leq (p + \varepsilon) - (p - 3\varepsilon) = 4\varepsilon$. $\square$

In what follows, we will rearrange or merge convergent series, or regroup terms, without further comment: all these are justified by Theorem 17 since the terms will be non-negative. It's also going to be convenient to rephrase Lemma 33 slightly: it says $\mu_0^*(S) + \mu_0^*(S^c) \geq 1$ holds for all sets S, and so $\mu_{0*}(S) = \mu_0^*(S)$ holds if and only if the reverse inequality $\mu_0^*(S) + \mu_0^*(S^c) \leq 1$ also holds. It will be a bit prettier to go for this 'reverse inequality' in what follows.

Theorem 36 (Carathéodory extension theorem, probability measures)**.** *Suppose $\mathcal{A}$ is an algebra on X and μ_0 is a pre-measure on $\mathcal{A}$ with $\mu_0(X) = 1$. Then $\mathcal{F} = \{S \subseteq X : \mu_{0*}(S) = \mu_0^*(S)\}$ is a σ-algebra containing $\mathcal{A}$, the restriction μ of μ_{0*} to $\mathcal{F}$ is a measure, and the restriction $\mu|_{\mathcal{A}}$ is μ_0. Moreover, μ is the unique measure on $\mathcal{F}$ such that $\mu|_{\mathcal{A}} = \mu_0$.*

Proof. It is trivial that if $F \in \mathcal{A}$ then $F \in \mathcal{F}$ and $\mu(F) = \mu_0(F)$. Indeed, F is a countable cover of F, and F^c is a countable cover of F^c, which verifies $\mu_0(F) \leq \mu_{0*}(F)$ and $\mu_0^*(F) \leq \mu_0(F)$. The middle inequality $\mu_{0*}(F) \leq \mu_0^*(F)$ is Lemma 33.

We first prove $\mathcal{F}$ is an algebra, and check that μ is additive on it. That is, we want to show $\mathcal{F}$ is closed under complements, finite intersection, and finite disjoint union. In each case, we'll use the same strategy we saw from Lemma 35 of picking 'good' countable covers and doing some set theory.

Exercise 18. *Prove that $\mathcal{F}$ is closed under complements, and $\mu(F) = 1 - \mu(F^c)$.*

Solution to Exercise 18:

Given $F \in \mathcal{F}$ and $\varepsilon > 0$, by definition of $\mu_0^*(F)$ there is a countable covering of F with size at most $\mu(F)+\varepsilon$. Similarly there is a countable covering of F^c with size at most $1-\mu(F)+\varepsilon$. By definition, these respectively witness $\mu_{0*}(F^c) \geq 1 - \mu(F) - \varepsilon$ and $\mu_0^*(F^c) \leq 1 - \mu(F) + \varepsilon$. Since this is true for all $\varepsilon > 0$, we have $\mu_{0*}(F^c) \geq 1 - \mu(F)$ and $\mu_0^*(F^c) \leq 1 - \mu(F)$. Now Lemma 33 gives us the middle inequality of

$$1 - \mu(F) \leq \mu_{0*}(F^c) \leq \mu_0^*(F^c) \leq 1 - \mu(F)\,,$$

which shows $F^c \in \mathcal{F}$ and $\mu(F^c) = 1 - \mu(F)$.

Dealing with intersections is a bit harder. Given $F, F' \in \mathcal{F}$ and $\varepsilon > 0$, choose disjoint ε-good countable coverings $A, A_1, \ldots$ of F and $B, B_1, \ldots$ of F^c of size at most respectively $\mu(F)+\varepsilon$ and $1-\mu(F)+\varepsilon$. We define $A', A'_1, \ldots$ and $B', B'_1, \ldots$ similarly with respect to F'.

The idea we should have is this: F is roughly A and F^c is roughly B, and the same for the prime versions. What if we have the very special situation that these 'roughly's were exactly true? Well, then $F \cap F'$ would be equal to $A \cap A'$, and $(F \cap F')^c = F^c \cup F'^c$ would be equal to $B \cup B'$; in particular, these two sets are countable covers (with 1 set!) of $F \cap F'$ and $(F \cap F')^c$ respectively. And in this case, those two sets would be disjoint and they would be contained in X (actually, they would cover X, but we don't need that). So we would have $\mu_0(A \cap A') + \mu_0(B \cup B') \leq 1$, and (because the two sets are countable covers) $\mu_0^*(F \cap F') + \mu_)^*\big((F \cap F')^c\big) \leq 1$. This, together with the reverse inequality from Lemma 33, gives us $\mu_0^*(F \cap F') = \mu_{0*}(F \cap F')$, which is what we want. All we've done here is a bit of basic set manipulation and using the definitions—draw a picture if you found it hard to follow!

Of course, the 'roughly's are not exact. We are told that $A, A_1, \ldots$ is a countable cover of F with size close to $\mu_0^*(F)$: the leftover is small but not empty, and the union $A \cup A_1 \cup \ldots$ covers F but might contain some more points too. Let's build up to this gently, by first considering the following special case.

Suppose all four leftovers consist only of empty sets, i.e. A is a covering of F, and so on. We still have that $A \cap A'$ is a covering of $F \cap F'$, and $B \cup B'$ is a covering of $(F \cap F')^c$.

The difference is $A \cap A'$ and $B \cup B'$ are no longer known to be disjoint: but $A \cap A'$ and $(B \cup B') \setminus (A \cap A')$ are. So we have

$$\mu_0(A \cap A') + \mu_0\big((B \cup B') \setminus (A \cap A')\big) \leq 1\,.$$

The 'error' here is $(B \cup B') \cap (A \cap A')$. This set is contained in $(B \cap A) \cup (B' \cap A')$, and by the union bound and Lemma 35, the measure of the error is at most

$$\mu_0\big((B \cap A) \cup (B' \cap A')\big) \leq \mu_0\big((B \cap A)\big) + \mu_0\big((B' \cap A')\big) \leq 8\varepsilon\,.$$

Putting these two inequalities together we get

$$\mu_0(A \cap A') + \mu_0(B \cup B') \leq 1 + 8\varepsilon\,.$$

As before this gives us

$$\mu_0^*(F \cap F') + \mu_0^*\big((F \cap F')^c\big) \leq 1 + 8\varepsilon\,.$$

Since this inequality holds for all $\varepsilon > 0$ we conclude $\mu_0^*(F \cap F') + \mu_0^*\big((F \cap F')^c\big) \leq 1$. The reverse inequality is Lemma 33, so $\mu_{0*}(F \cap F') = \mu_0^*(F \cap F')$ as desired.

The idea of the above special case was still the same idea: we'll write down a finite collection of disjoint sets in $\mathcal{A}$ and use the fact that their total pre-measure is at most 1. This time we needed to do a bit more work to argue that 'errors' are small in order to get to our inequality $\mu_0^*(F \cap F') \leq \mu_{0*}(F \cap F')$.

We now go through the general case, when the leftovers are not necessarily empty. The idea is to copy the above but 'fix' it to keep track of the new errors coming from the leftovers.

This is not all that difficult. The finite set theory is still the same, just that we need to throw a few more sets in to make our coverings.

First, $A \cap A', A_1, A_1', A_2, \ldots$ is a countable covering of $F \cap F'$: if x is in both F and F', and it's not in any A_i or A_i', then it must be in A and A' by definition of covering. Its size is at most $\mu_0(A \cap A') + 2\varepsilon$, so $\mu_0^*(F \cap F') \leq \mu_0(A \cap A') + 2\varepsilon$. And similarly $B \cup B', B_1, B_1', \ldots$ is a countable covering of $(F \cap F')^c$, whose size is at most $\mu_0(B \cup B') + 2\varepsilon$.

We still obtain, by the identical argument as above, $\mu_0(A \cap A') + \mu_0(B \cup B') \leq 1 + 8\varepsilon$. So

$$\mu_0^*(F \cap F') + \mu_0^*\big((F \cap F')^c\big) \leq 1 + 2\varepsilon + 2\varepsilon + 8\varepsilon = 1 + 12\varepsilon\,.$$

The proof now completes as before: since this holds for all $\varepsilon \geq 0$ we get $\mu_0^*(F \cap F') + \mu_0^*\big((F \cap F')^c\big) \leq 1$; together with Lemma 33 we get $\mu_{0*}(F \cap F') = \mu_0^*(F \cap F')$ as desired.

Exercise 19. *Prove that $\mathcal{F}$ is closed under disjoint union, and that μ is additive. That is, given disjoint sets $F, F' \in \mathcal{F}$, prove that $F \cup F' \in \mathcal{F}$ and $\mu(F \cup F') = \mu(F) + \mu(F')$. Hint: as with intersection, choose ε-good countable coverings for both the inner and outer pre-measure of F and of F'. First try to prove the special case where the leftovers are all empty, and use this to help you complete the general case. If you find the special case difficult, then try the very special case where the leftovers are empty and each main set is equal to the set it's covering first.*

Solution to Exercise 19:

As before, we choose near-optimal ε-good countable covers of F, F^c, F', F'^c and give them the same names as before ($A, A_1, \ldots$ covers F, and so on). Here we know what the 'right size' we are aiming for is: $\mu(F) + \mu(F')$.

The 'very special basic set theory' version of this is: $F \cup F'$ is $A \cup A'$ (which is a disjoint union), whose size is $\mu_0(A) + \mu_0(A')$, so $\mu_0^*(F \cup F') \leq \mu(F) + \mu(F')$. The complement $(F \cup F')^c$ is $B \cap B'$. Since these are disjoint sets we get $\mu_0(A) + \mu_0(A') + \mu_0(B \cap B') \leq 1$. As before, this inequality together with Lemma 33 tells us $F \cup F' \in \mathcal{F}$, and it also tells us that in fact $\mu(F \cup F') = \mu(F) + \mu(F')$: if not, we would have $\mu_0^*(F \cup F') < \mu_{0*}(F \cup F')$, in contradiction to Lemma 33.

Let's now do the 'special case' where A, A', B, B' on their own are coverings (i.e. the leftovers are empty) but they are not necessarily equal to the sets they cover.

We still want to use the same covering A, A' of $F \cup F'$. This covering immediately says $\mu_0^*(F \cup F') \leq \mu(F) + \mu(F') + 2\varepsilon$, and since that holds for all $\varepsilon > 0$ we get $\mu_0^*(F \cup F') \leq \mu(F) + \mu(F')$.

But now we cannot say that A and A' are disjoint: $A \cap A'$ might be non-empty; it might even (this is a 'bad case') potentially have large pre-measure. What we *can* say is that $A \cap B'$ is a covering of F: here we are using that F and F' are disjoint, so $F \subseteq F'^c \subseteq B'$.

Now Lemma 35 tells us $\mu_0(A' \cap B') \leq 4\varepsilon$. We can write $A = (A \setminus A') \cup (A \cap A')$, which is a disjoint union, and $A \cap B'$ is therefore a subset of $(A \setminus A') \cup (A \cap A' \cap B')$, which is also a disjoint union. So

$$\mu(A) - \mu(A \cap B') \geq \mu(A \cap A') - \mu(A \cap A' \cap B') \geq \mu(A \cap A') - 4\varepsilon\,.$$

In particular, if $\mu_0(A \cap A') > 5\varepsilon$, then $\mu_0(A \cap B') < \mu_0(A) - \varepsilon$. But this cannot be: then $A \cap B'$ is a covering of F whose size is strictly less than $\mu(F) + \varepsilon - \varepsilon = \mu(F)$, contradicting the definition of $\mu(F) = \mu_0^*(F)$. What this says is: the 'bad case' cannot occur; $\mu_0(A \cap A') \leq 5\varepsilon$.

We can now go on more or less as in the intersection case: $A, A' \setminus A, (B \cap B') \setminus (A \cup A')$ are disjoint sets, so their pre-measures sum to at most 1. Adding in the 'error's, we get

$$\mu_0(A) + \mu_0(A') + \mu_0(B \cap B') \leq 1 + 5\varepsilon + 8\varepsilon$$

where the 5ε is the $\mu_0(A \cap A')$ 'error', and the 8ε is the 'error' from $(B \cap B') \cap (A \cup A')$ which is contained in $(A \cap B) \cup (A' \cap B')$ as in the intersections proof.

Substituting in $\mu_0(A) \geq \mu(F)$ and $\mu_0(A') \geq \mu(F')$, and rearranging, we have $\mu_0(B \cap B') \leq 1 + 13\varepsilon - \mu(F) - \mu(F')$, which (since it holds for all positive ε) gives $\mu_0^*((F \cup F)^c) \leq 1 - \mu(F) - \mu(F')$. With Lemma 33 that proves $F \cup F' \in \mathcal{F}$; and furthermore (just as before) by Lemma 33 we have $\mu(F \cup F') = \mu(F) + \mu(F')$.

Finally, let's go to the general case, where we may have leftovers.

The first part is still easy: $A, A', A_1, A_1', \ldots$ is a countable covering of $F \cup F'$, and its size is at most $\mu(A) + \mu(A') + 2\varepsilon \leq \mu(F) + \varepsilon + \mu(F') + \varepsilon + 2\varepsilon = \mu(F) + \mu(F') + 4\varepsilon$. Since this holds for all $\varepsilon > 0$ we get $\mu_0^*(F \cup F') \leq \mu(F) + \mu(F')$.

As before, we should be concerned about the 'bad case' that $\mu_0(A \cup A')$ is large. This time, suppose $\mu_0(A \cup A') > 6\varepsilon$. Just as before, we have $\mu_0(A' \cap B') \leq 4\varepsilon$ by Lemma 33; and $\sum_{n\in\mathbb{N}} \mu_0(A' \cap B'_n) \leq \sum_{n\in\mathbb{N}} \mu_0(B'_n) \leq \varepsilon$ by ε-goodness. So if we take the countable covering

$$A \cap B', A_1, A \cap B'_1, A_2, A \cap B'_2, A_3, \ldots$$

of F (again, here we used that $B', B'_1, \ldots$ is a covering of F'^c which contains F), this countable covering has size strictly less than $\mu(F) + \varepsilon - (6\varepsilon - 4\varepsilon - \varepsilon) = \mu(F)$, in contradiction to the definition of $\mu(F)$.

Now we are more or less done as before: $A, A' \setminus A, (B \cap B') \setminus (A \cup A')$ are disjoint sets, so the pre-measures sum to at most 1. Adding in the 'errors' we have

$$\mu_0(A) + \mu_0(A') + \mu_0(B \cap B') \leq 1 + 6\varepsilon + 8\varepsilon$$

just as before, except that we have a slightly worse upper bound on $\mu_0(A \cap A')$. Substituting in $\mu_0(A) \geq \mu(F)$ and $\mu_0(A') \geq \mu(F')$, we get $\mu_0(B \cap B') \leq 1 - \mu(F) - \mu(F') + 14\varepsilon$.

Now $B \cap B', B_1, B'_1, \ldots$ is a countable covering of $(F \cup F')^c$ whose size is at most $1 - \mu(F) - \mu(F') + 16\varepsilon$. Since we can obtain this bound for any $\varepsilon > 0$ we conclude $\mu_0^*((F \cup F)^c) \leq 1 - \mu(F) - \mu(F')$.

Finally the proof now completes just as in the special case: Lemma 33 gives us the reverse inequality, so we have $F \cup F' \in \mathcal{F}$, and furthermore $\mu(F \cup F') = \mu(F) + \mu(F')$.

At this point we have verified that $(X, \mathcal{F}, \mu)$ is a discrete probability space: μ is finitely additive on the algebra $\mathcal{F}$. What remains is to show that $\mathcal{F}$ is also closed under countable disjoint unions and countable additivity. Suppose that $F_1, \ldots$ are a (countable) collection of disjoint sets in $\mathcal{F}$. At this point we will use a trick to simplify the problem using what we already proved: this is the same 'Littlewood philosophy' we saw a couple of times, of thinking of our collection of disjoint sets as the 'main' part which is a finite union together with the 'leftover' which is an infinite collection of sets but whose total size is small.

Consider the series $\sum_{i=1}^{\infty} \mu(F_i)$. If this series is divergent, there is some n such that $\mu\big(\bigcup_{i=1}^n F_i\big) = \sum_{i=1}^n \mu(F_i) > 1$, but this is impossible by Exercise 10(a) since $\mu(X) = 1$ and the union is a subset of X. So the series converges. Given any $\varepsilon > 0$ there exists n such that $\sum_{i=n+1}^{\infty} \mu(F_i) < \varepsilon$. The countable collection of sets $F_1 \cup \cdots \cup F_n, F_{n+1}, F_{n+2}, \ldots$ has the same union as our original collection of sets, its first element is in $\mathcal{F}$ as a finite union, and we have $\mu(F_1 \cup \cdots \cup F_n) = \mu(F_1) + \cdots + \mu(F_n)$ by finite additivity.

It's easy to find a countable covering of $F_1 \cup \ldots$ of the right size: we can just take the union of the coverings of each F_i: here we need the summable errors trick to make sure the total error is small. The reason for rearranging our problem by writing the union of $F_1 \cup \cdots \cup F_n$ and the remaining sets is that the complement of $F_1 \cup \ldots$ is a subset of the complement of $F_1 \cup \cdots \cup F_n$, and we know a countable covering of the latter of the right size exists: we can just reuse it. We'll now formalise this—most of the effort is to handle the notation.

Let $A_{11}, A_{12}, \ldots$ be a countable covering of $\bigcup_{i=1}^n F_i$ with size at most $\sum_{i=1}^n \mu(F_i) + \frac{1}{2}\varepsilon$, and for each $j \geq 2$, let $A_{j1}, A_{j2}, \ldots$ be a countable covering of F_{j-1+n} whose size is at most $\mu(F_{j-1+n}) + 2^{-j}\varepsilon$. Let $B_1, B_2, \ldots$ be a countable covering of $\Big(\bigcup_{i=1}^n F_i\Big)^c$ whose size is at most $1 + \varepsilon - \sum_{i=1}^n \mu(F_i)$.

For the countable cover of $\bigcup_{i=1}^\infty F_i$, we use the same approach as for the disjoint union of two sets: put all our covers together. We just need to keep track of the errors.

Using Theorem 17 the (diagonal enumeration) sequence

$$A_{11}, A_{21}, A_{12}, A_{31}, A_{22}, A_{13}, \dots$$

is a countable cover of $\bigcup_{i=1}^\infty F_i$, and its size is at most

$$\begin{aligned}&\sum_{i=1}^n \mu(F_i) + \tfrac{1}{2}\varepsilon + \sum_{i=2}^\infty \mu(F_{i-1+n}) + 2^{-i}\varepsilon \\ =&\sum_{i=1}^n \mu(F_i) + \sum_{j=n+1}^\infty \mu(F_j) + \varepsilon \sum_{i=1}^\infty 2^{-i} \le \sum_{i=1}^n \mu(F_i) + \varepsilon + \varepsilon = \sum_{i=1}^n \mu(F_i) + 2\varepsilon\,.\end{aligned}$$

This witnesses $\mu_0^*\big(\bigcup_{i=1}^\infty F_i\big) \le \sum_{i=1}^n \mu(F_i) + 2\varepsilon$.

The countable cover of $\Big(\bigcup_{i=1}^\infty F_i\Big)^c$ is easier. Since $\bigcup_{i=1}^n F_i$ is a subset of $\bigcup_{i=1}^\infty F_i$, it follows $\big(\bigcup_{i=1}^n F_i\big)^c$ is a superset of $\big(\bigcup_{i=1}^\infty F_i\big)^c$, so $B_1, B_2, \dots$ is a countable cover of $\big(\bigcup_{i=1}^\infty F_i\big)^c$. This witnesses $\mu_{0*}\big(\bigcup_{i=1}^\infty F_i\big) \ge -\varepsilon + \sum_{i=1}^n \mu(F_i)$.

By definition of n, we have $\sum_{i=1}^n \mu(F_i) \le \sum_{i=1}^\infty \mu(F_i) \le \varepsilon + \sum_{i=1}^n \mu(F_i)$. Thus for every $\varepsilon > 0$ there is an interval of length at most 3ε which contains $\mu_{0*}\big(\bigcup_{i=1}^\infty F_i\big)$, and $\sum_{i=1}^\infty \mu(F_i)$, and $\mu_0^*\big(\bigcup_{i=1}^\infty F_i\big)$. It follows these three numbers are all equal, proving $\bigcup_{i=1}^\infty F_i \in \mathcal{F}$ and the desired countable additivity.

We have now verified that $\mathcal{F}$ is a σ-algebra containing $\mathcal{A}$, and μ is a measure on it whose restriction to $\mathcal{A}$ is μ_0. Suppose μ' is any measure on $\mathcal{F}$ whose restriction to $\mathcal{A}$ is μ_0. If there exists a set $S \in \mathcal{F}$ such that $\mu'(S) \ne \mu(S)$, then either $\mu'(S) > \mu_0^*(S)$ or $\mu'(S^c) > \mu_0^*(S^c)$. By symmetry, we assume it is the former. By definition, there is a countable covering $A_1, \dots$ of S with elements from $\mathcal{A}$ whose size is strictly smaller than $\mu'(S)$. By the union bound, the union T of this countable covering has

$$\mu'(T) \le \sum_{i=1}^\infty \mu'(A_i) = \sum_{i=1}^\infty \mu_0(A_i) < \mu'(S)\,.$$

But this contradicts Exercise 10(a), since $S \subseteq T$. □

If you think hard about this proof, you'll notice several places earlier where the union bound would have made things easier, not just proving uniqueness at the end. Unfortunately, in those places we didn't yet know that μ is a measure, and we therefore weren't allowed to use the union bound.

The $\mathcal{F}$ to which we extended $\mathcal{A}$ is not unique: the uniqueness is that, for the particular $\mathcal{F}$ returned by Theorem 36, we don't have a choice about the function μ. A 'good' example of this is that we could have chosen to extend the algebra given by half-open intervals in $[0,1)$ to the Borel or to the Lebesgue σ-algebra on $[0,1)$, and we claimed those are different: but either way we don't have a choice about the measure.

However, here's a 'better' example in that it doesn't rely on our (here unproved) claim that the Borel and Lebesgue σ-algebras are different.

Exercise 20. *Let $X = [0,1)$, and let $\mathcal{A}$ consist of all subsets of X which are either countable or whose complement is countable. Define μ_0 on $\mathcal{A}$ by $\mu_0(A) = 0$ if A is countable, and $\mu_0(A) = 1$ if A^c is countable. Show that $\mathcal{A}$ is an algebra and μ_0 a pre-measure.*

Prove that there exists a σ-algebra $\mathcal{F}$ containing $\mathcal{A}$ and two different measures μ, μ' on $\mathcal{F}$, both of whose restrictions to $\mathcal{A}$ are μ_0. Hint: you may want to read to the end of these notes before attempting this.

Solution to Exercise 20:

We take $\mathcal{F}$ to be the Lebesgue σ-algebra $\bar{\mathcal{B}}([0,1))$, which we proved indeed contains $\mathcal{A}$. Now pick any two reasonably different functions $F, F^* : [0,1) \to [0,1]$ which are increasing, with $F(0) = F^*(0) = 0$ and with $F(x)$ and $F^*(x)$ both tending to 1 as $x \to 1$. For example $F(x) = x$ and $F^*(x) = x^2$.

We show below (Theorem 40 that F gives us a pre-measure on the algebra of half-open intervals, which by Theorem 36 extends to (the Lebesgue) measure λ on $\mathcal{L}([0,1))$ satisfying $\lambda([a,b)) = F(b) - F(a) = b - a$ for all $0 \le a < b \le 1$. Similarly, F^* gives us a pre-measure on the same algebra, which again extends to a measure μ on $\mathcal{L}([0,1))$, but this time satisfying $\mu([a,b)) = F^*(b) - F^*(a) = b^2 - a^2$ for all $0 \le a < b \le 1$. These are clearly two different measures on the same σ-algebra, because they for instance assign different measure to $[0, \frac{1}{2})$. But both (as is easy to check) assign measure 0 to all countable sets and hence 1 to all complements of countable sets: that is, they both extend μ_0.

That's all the exercise asks: but a couple of comments.

Applying Theorem 36 to μ_0 and $\mathcal{A}$ will return $\mathcal{A}$ and μ_0. The set is already a σ-algebra and μ_0 is already a measure: following the construction, if S is any set which is not countable then its outer pre-measure is 1 (even a countable union of countable sets cannot cover it!), while if S^c is not countable similarly S has inner pre-measure 0. So the only sets which have matching inner and outer pre-measure are the sets already in $\mathcal{A}$.

Formally we did not prove that Theorem 36 applied to the same algebra of half-open intervals with two different pre-measures will give the same σ-algebra both times (or that what we get will be specifically $\mathcal{L}([0,1))$): and indeed that's not always true. It is true in this case (why?), but if this disturbs you, then instead restrict the returned measures to the Borel σ-algebra $\mathcal{B}([0,1))$. Here we still get the same conclusion that the measures are different, and we can be sure that Theorem 36 will return us in both cases a σ-algebra containing all the half-open intervals (that's the conclusion of the theorem) and hence in both cases the returned σ-algebra must contain the σ-closure $\mathcal{B}([0,1))$.

We finish off this section by giving a 'short' proof of Theorem 36. This proof uses basically the same idea as the proof above, but we skip the intuition and the (unnecessary) proof of closure under intersection: one can do the proof of closure under any finite union (not necessarily disjoint) and quickly deduce additivity when the union happens to be disjoint. I wanted to take time to do one proof in lecture and let you think about something related for an exercise to better understand the proof: but logically we don't need to do both closure under intersection and union, and our proof of closure under disjoint union basically also worked for any union of two sets even if not disjoint.

Proof of Theorem 36. By Lemma 33, for any $S \subseteq X$ we have $\mu_0^*(S) + \mu_0^*(S^c) \ge 1$. Thus we can say $S \in \mathcal{F}$ if and only if $\mu_0^*(S) + \mu_0^*(S^c) \le 1$, and furthermore if we have countable

coverings $A_1, \dots$ of S and $B_1, \dots$ of S^c whose total size is at most $1+\rho$ (for any $\rho \geq 0$), then $\mu_0^*(S)$ is between $\sum_{n\in\mathbb{N}} \mu_0(A_n) - \rho$ and $\sum_{n\in\mathbb{N}} \mu_0(A_n)$.

It follows that for any $A \in \mathcal{A}$ we have $A \in \mathcal{F}$ and $\mu_0^*(A) = \mu(A) = \mu_0(A)$: A covers A and A^c covers A^c, with $\mu_0(A) + \mu_0(A^c) = 1$.

Suppose $F \in \mathcal{F}$. Then $\mu_0^*(F) + \mu_0^*(F^c) \leq 1$, which immediately says $F^c \in \mathcal{F}$.

Given $F, F' \in \mathcal{F}$, let $A, A_1, \dots$ be an ε-good countable covering of F with size at most $\mu(F) + \varepsilon$; let $B, B_1, \dots$ be similarly an ε-good countable covering of F^c with size at most $1 - \mu(F) + \varepsilon$, and define $A', A_1', \dots$ and $B', B_1', \dots$ similarly with respect to F'. Thn $A \cup A', A_1, A_1', \dots$ is a countable covering of $F \cup F'$ of size at most $\mu_0(A \cup A') + 2\varepsilon \geq \mu_0^*(F \cup F')$, and $B \cap B', B_1, B_1', \dots$ is a countable covering of $(F \cup F')^c$ of size at most $\mu_0(B \cap B') + 2\varepsilon \geq \mu_0^*((F \cup F')^c)$.

We have $\mu_0(A \cup A') + \mu_0\big((A \cup A') \setminus (B \cap B')\big) \leq 1$, and by Lemma 35 we have $\mu_0(A \cap B), \mu_0(A' \cap B') \leq 4\varepsilon$. It follows $\mu_0(A \cup A') + \mu_0(B \cap B') \leq 1 + 8\varepsilon$, and putting these inequalities together we have $\mu_0^*(F \cup F') + \mu_0^*\big((F \cup F')^c\big) \leq 1 + 12\varepsilon$. We also obtain $\mu_0^*(F \cup F')$ is between $\mu_0(A \cup A') - 12\varepsilon$ and $\mu_0(A \cup A') + 2\varepsilon$.

Suppose now in the above setting additionally F and F' are disjoint. We have $\mu_0(A \cup A') \leq \mu_0(A) + \mu_0(A')$ by the union bound. We claim $\mu_0(A \cap A') \leq 7\varepsilon$. Indeed, suppose this were false. By Lemma 35 we have $\mu_0(A' \cap B') \leq 4\varepsilon$, and we have $F \subseteq (F')^c \subseteq B \cup B_1 \cup \dots$.

We compare the following two countable coverings of F:

$$A \setminus A', A \cap A', A_1, A_2, \dots \quad \text{and} \quad A \setminus A', A \cap A' \cap B', A_1, B_1', \dots .$$

The former has by assumption size at least $\mu_0(A \setminus A') + \mu_0(A \cap A') \leq \mu(F) + \varepsilon$. The latter has size at most $\mu_0(A \setminus A') + \mu_0(A \cap A' \cap B') + 2\varepsilon \leq \mu_0(A \setminus A') + 6\varepsilon$. Writing these two out we get

$$\mu(F) + \varepsilon \geq \mu_0(A \setminus A') + \mu_0(A \cap A') \quad \text{and} \quad \mu(F) \leq \mu_0(A \setminus A') + 6\varepsilon\,,$$

where the second inequality records that the 'new' covering of F cannot has size smaller than $\mu(F)$, by definition of $\mu(F) = \mu_0^*(F)$. These two inequalities say $\mu(A \cap A') \leq 7\varepsilon$, as desired.

Thus $\mu_0(A \cup A') = \mu_0(A) + \mu_0(A') - \mu_0(A \cap A') \geq \mu(F) - \varepsilon + \mu(F) - \varepsilon - 7\varepsilon = \mu(F) + \mu(F') - 9\varepsilon$, and plugging this in to the bound previously we have

$$\mu(F) + \mu(F') - 21\varepsilon \leq \mu_0^*(F \cup F') \leq \mu(F) + \mu(F') + 4\varepsilon\,.$$

Since this last pair of inequalities, and the previous $\mu_0^*(F \cup F') + \mu_0^*\big((F \cup F')^c\big) \leq 1 + 12\varepsilon$, hold for all $\varepsilon > 0$ we conclude $F \cup F' \in \mathcal{F}$ and $\mu(F \cup F') = \mu_0^*(F \cup F') = \mu(F) + \mu(F')$. At this point we have proved $\mathcal{F}$ is an algebra and μ is a discrete probability measure on it (i.e. it is finitely additive).

It remains to show closure under countable union and countable additivity. Given $F_1, \dots$ in $\mathcal{F}$, because $\mathcal{F}$ is an algebra it is enough to assume the given sets are pairwise disjoint.

Let for each $n \in \mathbb{N}$ the sets $A_{1n}, A_{2n}, \dots$ be a countable coverings of F_n whose size is at most $\mu(F_n) + 2^{-n}\varepsilon$. Because the countable union of countable sets is countable, and applying Theorem 17, putting together all these coverings we have a countable covering of $\bigcup_{n\in\mathbb{N}} F_n$ whose size is at most $\varepsilon + \sum_{n\in\mathbb{N}} \mu(F_n)$, so $\mu_0^*\big(\bigcup_{n\in\mathbb{N}} F_n\big) \leq \varepsilon + \sum_{n\in\mathbb{N}} \mu(F_n)$.

Since $\sum_{n\in\mathbb{N}} \mu(F_n)$ converges, for some N the Nth partial sum exceeds $\sum_{n\in\mathbb{N}} \mu(F_n) - \varepsilon$. Because $\mathcal{F}$ is an algebra, there is a countable covering $B_1 \dots$ of $(F_1 \cup \dots \cup F_N)^c$ whose size is at most $1 - \mu(F_1) - \dots - \mu(F_N) + \varepsilon$. This is also a countable covering of $\big(\bigcup_{n\in\mathbb{N}} F_n\big)^c$, so

$$\mu_0^*\Big(\big(\bigcup_{n\in\mathbb{N}} F_n\big)^c\Big) \le 1 - \mu(F_1) - \dots - \mu(F_N) + \varepsilon \le 1 + 2\varepsilon - \sum_{n\in\mathbb{N}} \mu(F_n)\,.$$

Since both inequalities hold for all $\varepsilon > 0$, we have

$$\mu_0^*\big(\bigcup_{n\in\mathbb{N}} F_n\big) \le \sum_{n\in\mathbb{N}} \mu(F_n) \quad \text{and} \quad \mu_0^*\Big(\big(\bigcup_{n\in\mathbb{N}} F_n\big)^c\Big) \le 1 - \sum_{n\in\mathbb{N}} \mu(F_n)\,,$$

and Lemma 35 gives that both inequalities are in fact equalities, which verifies $\bigcup_{n\in\mathbb{N}} F_n \in \mathcal{F}$ and countable additivity.

We finish off with uniqueness. Given any measure μ' on $\mathcal{F}$ extending μ_0, for any $F \in \mathcal{F}$ we have $\mu'(F) \le \mu_0^*(F)$ because the size of any countable covering of F bounds above $\mu'(F)$ by the union bound; similarly $\mu'(F^c) \le \mu_0^*(F^c)$. Since μ' is a measure, we have $\mu'(F) + \mu'(F^c) = \mu(X) = \mu_0(X) = 1$. Now since $F \in \mathcal{F}$, we have $\mu_0^*(F) + \mu_0^*(F^c) \le 1$, so these inequalities must all be equalities, verifying $\mu' = \mu$. □

Carathéodory Extension Theorem: σ-finite case

Given X, an algebra $\mathcal{A}$ on X and a pre-measure μ_0 on $\mathcal{A}$, we say the pre-measure is *σ-finite* if we can write $X = \bigcup_{i=1}^\infty A_i$ where each A_i is in $\mathcal{A}$ and has $\mu_0(A_i) < \infty$; in other words, there is a countable covering of X with sets of finite measure. As in Exercise 15, we can assume without loss of generality that the sets in the countable covering are disjoint.

In this course, we will only be interested in σ-finite pre-measures.

Theorem 37 (Carathéodory extension theorem, σ-finite case)**.** *Suppose $\mathcal{A}$ is an algebra on X and μ_0 is a σ-finite pre-measure on $\mathcal{A}$. Then there is $\mathcal{F}$ a σ-algebra containing $\mathcal{A}$ and a measure μ on $\mathcal{F}$ such that the restriction $\mu|_{\mathcal{A}}$ to $\mathcal{A}$ is μ_0. Moreover μ is the unique measure on $\mathcal{F}$ such that $\mu_{\mathcal{A}} = \mu_0$.*

Proof. If $\mu_0(X) = 0$, then $\mu_0(A) = 0$ for all $A \in \mathcal{A}$. We can take $\mathcal{F} = \mathcal{P}(X)$ and $\mu(F) = 0$ for all $F \in \mathcal{F}$: this is clearly a measure.

We now deal with the case $0 < \mu_0(X) < \infty$. Let $\mu_0(X) = q$, and define a new pre-measure $\mu_0'(A) = q^{-1}\mu_0(A)$ for all $A \in \mathcal{A}$. By Theorem 36, there is a σ-algebra $\mathcal{F}$ on X containing $\mathcal{A}$ and a unique extension μ' of μ_0 on $\mathcal{F}$. We set $\mu(F) = q\mu'(F)$ for each $F \in \mathcal{F}$. By Theorem 30, μ is a measure on $\mathcal{F}$ extending μ_0; if it were not unique, this would contradict uniqueness of μ'.

At this stage we proved the case of Theorem 37 for $\mu_0(X) < \infty$; we will use this in the remaining case $\mu_0(X) = \infty$. Let $Y_1, \dots$ be a countable covering of X with disjoint sets in $\mathcal{A}$ of finite measure.

For each $i \in \mathbb{N}$, let $\mathcal{A}_i$ consist of all elements of $\mathcal{A}$ contained in Y_i.

Exercise 21. *Show that $\mathcal{A}_i$ is an algebra on Y_i and the restriction $\mu_0|_{\mathcal{A}_i}$ is a pre-measure.*

Solution to Exercise 21:

Since $Y_i \in \mathcal{A}$, so for any $A \in \mathcal{A}_i \subseteq \mathcal{A}$ we have $Y_i \setminus A \in \mathcal{A}$ and so $Y_i \setminus A \in \mathcal{A}_i$. This checks closure under complements. Closure under union is trivial: if $A, A' \in \mathcal{A}_i$ then they are in $\mathcal{A}$, so $A \cup A' \in \mathcal{A}$, so $A \cup A' \in \mathcal{A}_i$. We also for the same reason get $\mu_i(A \cup A') = \mu_i(A) + \mu_i(A')$ if additionally A and A' are disjoint.

Finally, suppose $A_1, \dots \in \mathcal{A}_i$ satisfy $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A}_i$. Then

$$\sum_{n=1}^{\infty} \mu_i(A_n) = \sum_{n=1}^{\infty} \mu(A_n) = \mu\Big(\bigcup_{n\in\mathbb{N}} A_n\Big) = \mu_i\Big(\bigcup_{n\in\mathbb{N}} A_n\Big),$$

checking countable pre-additivity.

Let $\mathcal{F}_i$ be the σ-algebra containing $\mathcal{A}_i$, and μ_i the unique measure on $\mathcal{F}_i$ extending $\mu_0|_{\mathcal{A}_i}$, obtained from the the finite measure case.

We define

$$\mathcal{F} = \Big\{ \bigcup_{i\in\mathbb{N}} F_i \,:\, F_i \in \mathcal{F}_i \text{ for each } i \in \mathbb{N} \Big\}.$$

Exercise 22. *Prove that $\mathcal{F}$ is a σ-algebra.*

Solution to Exercise 22:

Given a collection $F_i \in \mathcal{F}_i$ for $i \in \mathbb{N}$ whose union is F, the union of the F_i^c is F^c. Similarly, given a countable collection of sets in F, their intersections with each Y_i form a countable collection of sets in $\mathcal{F}_i$ whose union is therefore also in $\mathcal{F}_i$, so the union of the original countable collection is by definition in $\mathcal{F}$.

We now define a measure μ on $\mathcal{F}$. Given $F \in \mathcal{F}$, let $F_i = F \cap Y_i$ for each $i \in \mathbb{N}$. We define

$$\mu(F) = \sum_{i=1}^{\infty} \mu_i(F_i),$$

where if a term in the series is ∞, or if not but the series diverges, we take $\mu(F) = \infty$.

Exercise 23. *Verify that μ is a measure.*

Solution to Exercise 23:

We trivially have $\mu(\emptyset) = 0$. For countable additivity, suppose $F^{(1)}, F^{(2)}, \dots$ are a countable collection of pairwise disjoint sets in $\mathcal{F}$ with union F, and let $F_i^{(j)} = F^{(j)} \cap Y_i$, and $F_i = F \cap Y_i$, for each $i \in \mathbb{N}$.

If any $\mu_i\big(F_i^{(j)}\big)$ is ∞, then $\mu(F^{(j)}) = \infty$ and $\mu(F) = \infty$ by definition, verifying countable additivity in this case. So suppose that all $\mu_i\big(F_i^{(j)}\big)$ are real numbers.

If $\sum_{i=1}^{\infty} \mu_i\big(F_i^{(j)}\big)$ does not converge for some j, then $\mu(F^{(j)}) = \infty$ by definition. Furthermore, $\mu_i(F_i) \geq \mu_i(F_i^{(j)})$ because μ_i is a measure, so by Theorem 19 also $\sum_{i=1}^{\infty} \mu_i(F_i)$ does not converge and $\mu(F) = \infty$, as required. So suppose these sums are all convergent.

Similarly, if $\sum_{j=1}^{\infty} \mu_i\big(F_i^{(j)}\big)$ does not converge for some i, then $\mu_i(F_i) = \infty$ and so $\mu(F) = \infty$. But again we have $0 \leq \mu_i(F_i^{(j)}) \leq \mu(F^{(j)})$, so $\sum_{j=1}^{\infty} F^{(j)})$ diverges, verifying countable additivity. So we suppose these sums are all convergent too.

Now if either of $\sum_{i=1}^{\infty}\sum_{j=1}^{\infty}\mu_i\big(F_i^{(j)}\big)$ and $\sum_{j=1}^{\infty}\sum_{i=1}^{\infty}\mu_i\big(F_i^{(j)}\big)$ diverges, by Theorem 17 so does the diagonal enumeration series and so does the other one: then (from the first sum) by definition $\mu(F)=\infty$ and (from the second) $\sum_{j=1}^{\infty}\mu(F^{(j)})$ diverges, again as required.

Finally suppose those sums too are convergent. Then by Theorem 17 they are equal, verifying countable additivity in this case too.

(The final case here is really 'the point'; the rest is just handling ∞.)

What remains to prove is that μ is unique. This is quite easy. For any $F\in\mathcal{F}$, we are required to have $\mu(F\cap Y_i)=\mu_i(F\cap Y_i)$, and $\mu_i(F\cap Y_i)$ is uniquely defined. Now any $F\in\mathcal{F}$ is the union of $F\cap Y_i$ for $i\in\mathbb{N}$: this is a countable disjoint union and because μ is a measure we necessarily have $\mu(F)=\sum_{i\in\mathbb{N}}\mu(F\cap Y_i)$. □

It's convenient to note that we proved a little more than the theorem states. The same argument shows:

Theorem 38. *Suppose for each $i\in\mathbb{N}$ we have a set Y_i, a σ-algebra $\mathcal{F}_i$ on it, and a measure μ_i on $\mathcal{F}_i$. Suppose that the Y_i are pairwise disjoint. Then there is a σ-algebra $\mathcal{F}$ on $\bigcup_{i\in\mathbb{N}}Y_i$ and a measure μ on $\mathcal{F}$, such that for each $i\in\mathbb{N}$ we have $\mathcal{F}_i=\{F\in\mathcal{F}:F\subseteq Y_i\}$ and $\mu|_{\mathcal{F}_i}=\mu_i$; furthermore for this $\mathcal{F}$ the measure μ is unique.*

Carathéodory Extension Theorem, general case

This part is really for interest only: we will not deal with non-σ-finite pre-measures in this course. We do not actually need any hypothesis about X: but we are then not able to prove that the measure we obtain is unique.

Theorem 39 (Carathéodory's Extension Theorem)**.** *Suppose $\mathcal{A}$ is an algebra on X and μ_0 is a pre-measure on $\mathcal{A}$. Then there is $\mathcal{F}$ a σ-algebra containing $\mathcal{A}$ and a measure μ on $\mathcal{F}$ such that the restriction $\mu|_{\mathcal{A}}$ to $\mathcal{A}$ is μ_0.*

One way to prove this is as follows.

Proof. Since we dealt with the σ-finite case, we now assume μ_0 is not σ-finite, i.e. any countable union of sets in $\mathcal{A}$ with finite measure is not equal to X. We will use this assumption in what follows.

Before we get on with the main proof, we'll need to deal with a 'concern' here. If $A_1,\dots$ and $A_1',\dots$ are countable collections of disjoint sets in $\mathcal{A}$, which happen to have the same union $\bigcup_{n\in\mathbb{N}}A_i=\bigcup_{n\in\mathbb{N}}A_i'=S$, what can we say about pre-measure?

If S is in $\mathcal{A}$, then countable pre-additivity says

$$\mu_0(S)=\sum_{n=1}^{\infty}\mu_0(A_n)=\sum_{n=1}^{\infty}\mu_0(A_n')\,,$$

and in particular the two sums agree.

But if S is *not* in $\mathcal{A}$, we can't apply countable pre-additivity. What if the 'bad situation' that the two sums disagree occurs? Certainly the Extension Theorem would be false: because

S will be in the returned $\mathcal{F}$ (countable union), so $\mu(F)$ must be defined and equal to both sums (countable additivity).

So, somehow, we must have excluded this possibility in the previous two proofs. It's not too obvious where we did so: it turns out we did it in Theorem 36 by proving countable additivity and using Lemma 33. If you apply this to the 'bad situation', you'll see (with some work!) it implies there is a countable disjoint covering of X with sets in $\mathcal{A}$ with size smaller than 1, which countable pre-additivity bans. And (again this needs a bit of thought) if we have such a contradiction in the setup of Theorem 37, then there must also be an example in one of the finite-measure pieces, which we already excluded.

But those arguments rely on our pre-measure being finite, or at least σ-finite. This one is not: we need to exclude the bad situation here. So suppose for a contradiction it occurs; suppose the sum $\sum_{n=1}^{\infty} \mu_0(A'_n)$ is larger (either it's a larger real number, or it does not converge or has a term ∞ but $\sum_{n=1}^{\infty} \mu_0(A_n)$ does converge).

Then there is some N such that $\sum_{n=1}^{N} \mu_0(A'_n) > \sum_{n=1}^{\infty} \mu_0(A_n)$. Let $A' = \bigcup_{n=1}^{N} A'_n$. The important point to note is A' is a finite union of sets in $\mathcal{A}$, so it's in $\mathcal{A}$ and we can try to use countable pre-additivity. Since $A' \subseteq \bigcup_{n\in\mathbb{N}} A_n$, we have

$$\bigcup_{n\in\mathbb{N}} (A_n \cap A') = A' \,,$$

and this is a countable disjoint union of sets in $\mathcal{A}$: by countable pre-additivity we have

$$\sum_{n=1}^{\infty} \mu_0(A_n \cap A') = \mu_0(A') \,.$$

But on the other hand we have $\sum_{n=1}^{\infty} \mu_0(A_n) < \mu_0(A')$ by assumption. Since $0 \le \mu_0(A_n \cap A') \le \mu_0(A)$, we have by Theorem 19

$$\mu_0(A') = \sum_{n=1}^{\infty} \mu_0(A_n \cap A') \le \sum_{n=1}^{\infty} \mu_0(A_n) < \mu_0(A') \,,$$

which is a contradiction. This proves the 'bad situation' cannot occur in any pre-measure.

With this in mind, let's get on with the proof.

Let $\{Y_i : i \in I\}$ be the collection of σ-finite subsets of X (i.e. subsets of X which can be written as a countable union of sets in $\mathcal{A}$ of finite measure). (We don't know I is countable here; it might well not be.)

Exercise 24. *Show that for each $i \in I$ there is an algebra $\mathcal{A}_i$ on Y_i containing $\{A \in \mathcal{A} : A \subseteq Y_i\}$, and a unique pre-measure μ_{0i} on $\mathcal{A}_i$ whose restriction to $\mathcal{A} \cap \mathcal{A}_i$ agrees with μ_0.*

Hint: what is missing is closure under complements. You will need to use countable pre-additivity to justify uniqueness of the pre-measure.

Solution to Exercise 24:

Let $\mathcal{A}_i = \big\{A, Y_i \setminus A \,:\, A \in \mathcal{A}\,,\, A \subseteq Y_i\big\}$. We claim that $\mathcal{A}_i$ is an algebra on Y_i. By definition it is closed under complements. Suppose $A, A' \in \mathcal{A}_i$. If $A, A' \in \mathcal{A}$, then $A \cup A' \in \mathcal{A}$

and hence $A \cup A' \in \mathcal{A}_i$. If this is not the case, then $Y_i \notin \mathcal{A}$ and we have $A' = Y_i \setminus B'$ for some $B' \in \mathcal{A} \cap \mathcal{A}_i$. If $A \in \mathcal{A}$, we can write

$$A \cup A' = A \cup (Y_i \setminus B') = Y_i \setminus (B' \setminus A)$$

which is in $\mathcal{A}_i$. Finally, if $A \notin \mathcal{A}$ then we have $A = Y_i \setminus B$ for some $B \in \mathcal{A} \cap \mathcal{A}_i$, and again

$$A \cup A' = (Y_i \setminus B) \cup (Y_i \setminus B') = Y_i \setminus (B \cap B')$$

which is in $\mathcal{A}_i$. This verifies $\mathcal{A}_i$ is an algebra on Y_i.

Observe that we can write any $A \in \mathcal{A}_i$ as a countable disjoint union of sets in $\mathcal{A}$: for sets which are in $\mathcal{A}$ this is obvious, while if $A = Y_i \setminus B$ for some $B \in \mathcal{A}$, we can write $Y_i = C_1 \cup \dots$ as a union of sets in $\mathcal{A}$ because Y_i is σ-finite; as usual without loss of generality we can assume that the union is a disjoint union, and then $C_n \setminus B$ is in $\mathcal{A}$ for each n and the (disjoint countable) union of the $C_n \setminus B$ is $Y_i \setminus B$ as required.

So let $A = B_1 \cup \dots$ be a countable disjoint union of sets in $\mathcal{A}$, and define $\mu_{0i}(A) = \sum_{n=1}^{\infty} \mu_0(B_n)$. This is well defined because while there might be two different ways to write A as a countable disjoint union of sets in $\mathcal{A}$, we proved above that the two series we get must be equal (the 'bad situation' does not occur).

We need to justify that this is really a pre-measure. For additivity, suppose this definition gives $\mu_{0i}(A \cup A') \neq \mu_{0i}(A) + \mu_{0i}(A')$ for some disjoint $A, A' \in \mathcal{A}_i$. Unpacking the definition, we obtain two series whose sum is different (we can merge the two series on the right to one by Theorem 17, being a little careful to handle divergence correctly) but which both represent a countable disjoint union of sets in $\mathcal{A}$ with the same union: this is the 'bad situation' we excluded. This proves additivity.

For countable pre-additivity, we use much the same argument, just that we have a countable collection of series. Theorem 17 still applies (though a little more care is needed to handle divergence correctly).

Finally, uniqueness is quite easy at this stage. Suppose μ_0' is any pre-measure on $\mathcal{A}_i$ whose restriction to $\mathcal{A} \cap \mathcal{A}_i$ agrees with μ_0. Given any set $A \in \mathcal{A}_i$, writing A as a countable disjoint union of sets $D_1, \dots$ in $\mathcal{A}$, by countable pre-additivity we have

$$\mu_{0i}(A) = \sum_{n=1}^{\infty} \mu_{0i}(D_n) = \sum_{n=1}^{\infty} \mu_0(D_n) = \sum_{n=1}^{\infty} \mu_0'(D_n) = \mu_0'(A)\,.$$

(It might look like we should've just used this argument from the beginning to justify we have a pre-measure. The point is we couldn't use countable pre-additivity on all of $\mathcal{A}_i$, only on $\mathcal{A}_i \cap \mathcal{A}$, until we proved that we have a pre-measure on $\mathcal{A}_i$.)

We can now apply the σ-finite case for each $i \in I$ to obtain σ-algebras $\mathcal{F}_i$ and unique measures μ_i.

Exercise 25. *Prove that $\mathcal{F}'$ consisting of all sets in $\bigcup_{i \in I} \mathcal{F}_i$ and their complements forms a σ-algebra.*

Hint: a countable union of countable sets is countable.

Solution to Exercise 25:

$\mathcal{F}'$ is obviously closed under complements, so what is left is to verify closure under countable union. A set in $\mathcal{F}'$ is either in one of our $\mathcal{F}_i$, or it is the complement of such a set.

First consider $A_1, \dots \in \bigcup_{i \in I} \mathcal{F}_i$. For each A_n, there exists $i \in I$ such that $A_n \in \mathcal{F}_i$, so there is a countable J such that all A_n are in some $\mathcal{F}_j$ with $j \in J$. But now $\bigcup_{j \in J} Y_j$ is a countable union of countably many sets of finite measure, so it is σ-finite, so it is Y_i for some fixed $i \in I$. Now $A_1, \dots \in \mathcal{F}_i$, which is a σ-algebra, so the union is also in $\mathcal{F}_i$ and hence in $\mathcal{F}'$.

The remaining case is rather similar: we can assume at least one set is a complement. Given $A_1, \dots, B_1, \dots \in \bigcup_{i \in I} \mathcal{F}_i$, by the same argument there is a single $\mathcal{F}_i$ containing all of those sets. Now the union

$$\bigcup_{n \in \mathbb{N}} A_n \cup \bigcup_{n \in \mathbb{N}} (X \setminus B_n) = X \setminus \Big(\bigcap_{n \in \mathbb{N}} B_n \setminus \bigcup_{n \in \mathbb{N}} A_n \Big)$$

is the complement of a set in $\mathcal{F}_i$, so it is in $\mathcal{F}'$.

We define μ' on $\mathcal{F}'$ as follows. If $F \in \mathcal{F}'$ is in $\mathcal{F}_i$ for some $i \in I$, let $\mu'(F) = \mu_i(F)$. If not, let $\mu(F) = \infty$.

Exercise 26. *Prove that μ' is well-defined and that it is a measure.*

Solution to Exercise 26:

We deal with well-definedness first. It's obvious that our formula assigns a value in $\mathbb{R}^{\infty}_{\geq 0}$ to every set in $\mathcal{F}'$: what we are worried about is the possibility it might try to assign two different values.

There is only one potential problem here. Suppose $F \in \mathcal{F}'$ is in some $\mathcal{F}_i$. It could also be in some other $\mathcal{F}_j$, and then we need to argue $\mu_i(F) = \mu_j(F)$. Look at the σ-algebra $\mathcal{F}_k$ on $Y_k = Y_i \cap Y_j$ (this is a σ-finite set: why?); this σ-algebra is $\mathcal{F}_i \cap \mathcal{F}_j$, and we proved μ_k is the unique measure on it which extends μ_{0k}. Now $\mu_i|_{\mathcal{F}_k}$ and $\mu_j|_{\mathcal{F}_k}$ are both measures extending μ_{0k}, so they must be identical, in particular $\mu_i(F) = \mu_j(F)$.

We now need to argue μ' is a measure. The trick here is to notice that $\bigcup_{i \in I} \mathcal{F}_i$ is closed under countable union, which we proved in Exercise 25, and μ' is countably additive on this collection of sets: again, because given any countable collection of disjoint sets in $\bigcup_{i \in I} \mathcal{F}_i$ there is a single $i \in I$ such that $\mathcal{F}_i$ contains all of them: the countable union is in $\mathcal{F}_i$ also, μ' and μ_i agree on all sets in $\mathcal{F}_i$, and μ_i is a measure so the countable additivity holds.

If we are given a collection of sets $F_1, \dots$ which is not in $\bigcup_{i \in I} \mathcal{F}_i$, then we need to argue their union is also not in $\bigcup_{i \in I} \mathcal{F}_i$ (so that both sides are assigned measure ∞). Here we use the fact that X is not σ-finite. If we have $F_1 \cup \dots = F$, where all sets are in $\mathcal{F}'$, at least one of the left side sets is not in $\bigcup_{i \in I} \mathcal{F}_i$, and the right side set is in $\bigcup_{i \in I} \mathcal{F}_i$, then we can rearrange the set equality to express X as a countable union of sets in $\bigcup_{i \in I} \mathcal{F}_i$ (using the same set identities as in Exercise 25). Now each set in $\bigcup_{i \in I} \mathcal{F}_i$ is a subset of a σ-finite set, so is itself σ-finite, and (again using that the countable union of countable sets is countable) we see X is σ-finite, a contradiction. We conclude that if a set on the left is not in $\bigcup_{i \in I} \mathcal{F}_i$, so F is also not in $\bigcup_{i \in I} \mathcal{F}_i$ and we have countable additivity with both sides evaluating to ∞.

This is not quite the final measure we need. Let $\mathcal{F} = \sigma\big(\mathcal{F}' \cup \mathcal{A}\big)$, and define μ by $\mu(F) = \mu'(F)$ if $F \in \mathcal{F}'$, and $\mu(F) = \infty$ otherwise. By definition $\mathcal{F}$ is a σ-algebra and

contains $\mathcal{A}$, and by definition $\mu(A) = \mu_0(A)$ for all $A \in \mathcal{A}$.

Exercise 27. *Prove that μ is a measure.*

Solution to Exercise 27:

What we have to prove is that μ satisfies countable additivity (since obviously $\mu(\emptyset) = 0$). So suppose we have $F_1, \dots \in \mathcal{F}$ with $\bigcup_{n\in\mathbb{N}} F_n = F$.

If all of $F_1, \dots$ are in $\mathcal{F}'$, then so is F and because μ' is a measure the countable additivity is satisfied for these sets.

So we may assume at least one F_j is not in $\mathcal{F}'$; this F_j has $\mu(F_j) = \infty$, and what we need now to show is $\mu(F) = \infty$. Suppose that were false: then $F \in \mathcal{F}'$ (because all sets in $\mathcal{F} \setminus \mathcal{F}'$ are assigned measure ∞). By definition, it follows $F \in \mathcal{F}_i$, which is a σ-algebra on Y_i, for some $i \in I$, and $Y_i = C_1 \cup \dots$ is a countable union of sets in $\mathcal{A}$ of finite measure.

Intuitively it is 'obvious' that F_j will be in the σ-algebra $\mathcal{G}$ on Y_i generated by $\{F \in \mathcal{F}' : F \subseteq Y_i\} \cup \mathcal{A}_i$ (and therefore in $\mathcal{F}'$, which is the desired contradiction). But this needs a small trick to prove.

Let $\mathcal{G}^* = \{S \subseteq X : S \cap Y_i \in \mathcal{G}\}$. It is easy to check that $\mathcal{G}^*$ is a σ-algebra (since $\mathcal{G}$ is). Now $\mathcal{G}^*$ contains all elements of $\mathcal{F}'$: by definition $Y_i \in \mathcal{F}'$ and so for any $F \in \mathcal{F}'$ we have $F \cap Y_i \in \mathcal{F}'$, so $F \cap Y_i \in \mathcal{G}$ and by definition $F \in \mathcal{G}^*$. Similarly, $\mathcal{G}^*$ contains all elements of $\mathcal{A}$: given $A \in \mathcal{A}$, for each $n \in \mathbb{N}$ we have $A \cap C_n \in \mathcal{A}$, and $A \cap C_n \in \mathcal{A}_i$ by definition, so each $A \cap C_n$ is in $\mathcal{G}$, and therefore their countable union $A \cap Y_i$ is also in $\mathcal{G}$ and by definition $A \in \mathcal{G}^*$.

Since $\mathcal{F}' \cup \mathcal{A} \subseteq \mathcal{G}^*$ we have $\mathcal{F} = \sigma(\mathcal{F}' \cup \mathcal{A}) \subseteq \sigma(\mathcal{G}^*) = \mathcal{G}^*$. Since $F_j \in \mathcal{F}$, it follows $F_j \in \mathcal{G}^*$ and so $F_j \cap Y_i = F_j \in \mathcal{G}$.[6]

Now we are done: $\mathcal{A}_i$ is contained in $\mathcal{F}_i$ which is contained in $\{F \in \mathcal{F}' : F \subseteq Y_i\}$ by definition. So $\mathcal{G} = \sigma\big(\{F \in \mathcal{F}' : F \subseteq Y_i\}\big) \subseteq \sigma(\mathcal{F}') = \mathcal{F}'$, and in particular $F_j \in \mathcal{F}'$, the desired contradiction.

□

A rather harder exercise is to identify where we lost uniqueness.

Exercise 28. *At what step in the above can we no longer say that our measures are unique?*

Solution to Exercise 28:

In a couple of places we assigned some sets measure ∞ 'arbitrarily'; we weren't forced to do so, and (sometimes) it would have been possible to assign some of these sets finite measure as well. I think this is true for the assignment of ∞ in μ' already: it is certainly true for the sets to which we assign measure ∞ when we go from μ' to μ.

The measure μ we constructed is sometimes called the 'maximal' extension of μ_0: all the sets with finite measure in μ must be assigned that measure in any extension (and this is true for many sets with measure ∞ too), so if μ^* is any other extension of μ_0 to $\mathcal{F}$, then we have $\mu^*(F) \leq \mu(F)$ for all $F \in \mathcal{F}$.

[6] It's tempting to say that we just copy the 'sequence' of actions that construct $F_j \in \mathcal{F}$ but intersect everything with Y_i. The problem is this is not really a 'sequence'; the idea can be formalised but it needs transfinite induction.

The proof we've just seen of the Carathéodory Extension Theorem is rather non-standard. The usual route goes by defining outer pre-measure as we did, then *Carathéodory measurable*, which is a condition that only needs to talk about the outer pre-measure. One then proves that the collection of Carathéodory measurable sets forms a σ-algebra and that μ_0^* is a measure on it. This approach is a bit shorter than the route we took, but it (a) requires you to understand the (notoriously non-intuitive) definition of Carathéodory measurable, and (b) it contains rather more magic tricks with sets.

A *ring* $\mathcal{R}$ on X is defined to be a collection of sets closed under finite union and set difference (i.e. if $A, B \in \mathcal{R}$ then $A \setminus B \in \mathcal{R}$).

It's easy to prove that $\mathcal{R} \cup \{X \setminus R : R \in \mathcal{R}\}$ is an algebra on X. While we won't need it (it's always going to be easy enough for us to construct pre-measures on algebras) there is an even stronger version of the Carathéodory Extension Theorem, which requires only a pre-measure on a ring to start with.

Exercise 29. *Suppose $\mathcal{R}$ is a ring on X, and μ_0 is a pre-measure on $\mathcal{R}$ (the same conditions as for on an algebra apply). Let $\mathcal{A} = \mathcal{R} \cup \{X \setminus R : R \in \mathcal{R}\}$. Prove that there is a pre-measure μ_0' on $\mathcal{A}$ with $\mu_0'|_{\mathcal{R}} = \mu_0$. Explain why if X is σ-finite the choice of μ_0' is unique.*

Solution to Exercise 29:

Our proof earlier that the 'bad situation' cannot occur does not require taking complements (you can do intersection via set difference), so it is valid here as well.

If X is σ-finite, then we can write $X = C_1 \cup \dots$ as a disjoint union of sets of finite measure. For any $R \in \mathcal{R}$, we can therefore write

$$X \setminus R = \bigcup_{n \in \mathbb{N}} (C_n \setminus R)$$

is a countable disjoint union of sets in $\mathcal{R}$, and we then set $\mu_0'(X \setminus R) = \sum_{n \in \mathbb{N}} \mu_0(C_n \setminus R)$. This is well-defined (i.e. does not depend on the choice of $C_1, \dots$) because the 'bad situation' does not occur: for the same reason μ_0' satisfies countable pre-additivity, and the uniqueness in this case is obvious (we had no choice but to make this definition, otherwise we would immediately break countable pre-additivity).

If X is not σ-finite, we do something rather similar. Given $R \in \mathcal{R}$, obviously we write $\mu_0'(R) = \mu_0(R)$. If $X \setminus R$ can be covered by a countable collection of finite pre-measure sets in $\mathcal{R}$, we write

$$\mu_0'(X \setminus R) = \inf_{\substack{F_1, \dots \\ \text{covering } X \setminus R}} \sum_{n \in \mathbb{N}} \mu_0(F_n)\,,$$

and if not then we write $\mu_0'(X \setminus R) = \infty$. Observe that we can (as usual) always assume we are given a countable covering with disjoint sets in $\mathcal{R}$. In addition, we can assume our countable covering has union equal to $X \setminus R$: given $R_1, \dots \in \mathcal{R}$, the sets $R_1 \setminus R, R_2 \setminus R$ and so on are also in $\mathcal{R}$.

Our definition of μ_0' is then well-defined because the 'bad situation' does not occur. We need to verify countable pre-additivity. Given $A_1, \dots, A \in \mathcal{A}$ such that $A = \bigcup_{i \in \mathbb{N}} A_i$ with $A_1, \dots$ disjoint, if each A_i has finite pre-measure then we have countable pre-additivity because otherwise the 'bad situation' occurs (here we use Theorem 17 to justify merging the

countably many series, and we assume the countable coverings are disjoint and the countable covering of each A_i has union A_i, so that all the sets in all countable coverings of $A_1, \ldots$ are pairwise disjoint). So there is some A_i whose pre-measure is not finite, and in order for there to be a contradiction to countable pre-additivity we have $\mu_0'(A) < \infty$.

But this is also impossible: since $\mu_0'(A) < \infty$, there is a countable covering of A with finite pre-measure sets in $\mathcal{R}$, the sum of whose pre-measures converges to a real number. But this countable covering is also a countable covering of A_i and this witnesses $\mu_0'(A_i) < \infty$.

This implies that we can replace 'algebra' with 'ring' in each of our statements of the Carathéodory Extension Theorem.

Constructing a pre-measure

We've now done the 'general measure theory'. Let's see one application: construction of measures on $[0, 1)$ and $\mathbb{R}$.

It's convenient, again, to start with $[0, 1)$. The algebra $\mathcal{A}$ we define on $[0, 1)$ is

$$\left\{ I_1 \cup \cdots \cup I_k \,:\, k \in \mathbb{N} \text{ and } I_i \text{ is a half-interval for each } 1 \leq i \leq k \right\}.$$

Here 'half-interval' means of the form $[a, b)$ with $0 \leq a \leq b \leq 1$. Observe that if two half-intervals have non-empty intersection or share an endpoint, their union is again a half-interval, so we can assume that $I_1 \cup \cdots \cup I_k$ is a disjoint union and no two half-intervals share an endpoint (they are *separated*). Note that any given set which is a finite disjoint union of half-intervals has a unique representation with separated half-intervals. The complement of such a set is transparently also a finite union of half-intervals, and the union of two such sets is even more transparently a union of finitely many half-intervals. This verifies that $\mathcal{A}$ is really an algebra.

We might as well (it won't make things harder) construct a general family of measures. Let $F : [0, 1) \to \mathbb{R}$ be any continuous increasing function with $F(0) = 0$, and for a half-interval I with endpoints $a < b$ let $\mu_0(I) = F(b) - F(a)$. We extend this to any finite separated union of half-intervals additively—note that this is required if we intend to define a measure at all.

For the Borel measure, we'd choose $F(x) = x$, so that $\mu_0([a, b]) = b - a$. In general, if $F(0) = 0$ and $F(1) = 1$ then F is the cumulative distribution function of a distribution on $[0, 1)$, and we're trying to formally construct that distribution as a measure.

Theorem 40. *For any increasing continuous $F : [0, 1) \to \mathbb{R}$, there is a unique pre-measure μ_0 on $\mathcal{A}$ such that $\mu_0([a, b)) = F(b) - F(a)$ for all $0 \leq a < b \leq 1$.*

Proof. By definition we have $\mu_0(\emptyset) = 0$. As usual, it's best to first show μ_0 is finitely additive, then try to prove countable pre-additivity.

For additivity, suppose $I_1 \cup \cdots \cup I_k$ and $I_1' \cup \cdots \cup I_{k'}'$ are in $\mathcal{A}$ and are disjoint. Then $I_1 \cup \cdots \cup I_k \cup I_1' \cup \cdots \cup I_{k'}'$ is their union: this is a finite disjoint half-interval representation and additivity follows, cancelling the finitely many terms where endpoints are shared to obtain the separated representation.

Given a countable collection of pairwise disjoint members $A_1, A_2, A_3, \ldots$ of $\mathcal{A}$ whose union is $A \in \mathcal{A}$, for each member of $\mathcal{A}$ fix a separated finite half-interval representation. Then $\mu_0(A_i)$ is equal to the sum of the measures of the half-intervals in the representation of A_i. Observe that the series $\sum_{i=1}^{\infty} \mu_0(A_i)$ has non-negative terms, so its partial sums are monotone increasing. If we replace in this series the term A_1 with a collection of terms, one for each half-interval in the representation of A_1, and so on for $A_2, \ldots$ (but we do not change the order) then our new series has partial sums which are also monotone increasing and a subsequence of which are the partial sums of $\sum_{i=1}^{\infty} \mu_0(A_i)$. It follows that either both series diverge or both series converge to the same limit. We can therefore without loss of generality assume that each A_i is in fact a single half-interval. Similarly, by taking intersections with each half-interval in the finite separated half-interval representation of A (and applying finite additivity if we succeed for each) it is enough to assume A is a single half-interval.

We first deal with divergence. If $\sum_{i=1}^{\infty} \mu_0(A_i)$ diverges, there exists n such that, letting $S = \bigcup_{i=1}^{n} A_i$, we have $F(1) - F(0) < \sum_{i=1}^{n} \mu_0(A_i) = \mu_0(S)$, by finite additivity. But this is impossible because then $\mu_0([0,1)) = \mu_0(S) + \mu_0(S^c) > F(1) - F(0) = \mu_0([0,1))$.

We can therefore assume the series converges. For any given n, we have $\bigcup_{i=1}^{n} A_i$ is a subset of A in $\mathcal{A}$, so $\sum_{i=1}^{n} \mu_0(A_i) \leq \mu_0(A)$ by finite additivity. Since the partial sums are bounded by $\mu_0(A)$, it follows that $\sum_{i=1}^{\infty} \mu_0(A_i) \leq \mu_0(A)$.

The inequality we need in the other direction is a good deal harder. Maybe the easiest route is to use Heine-Borel, Theorem 20, to reduce it to a finite problem.

Given $\varepsilon > 0$, for each $A_i = [a, b)$, by continuity of F we choose $a' < a$ such that $F(a) - F(a') < \varepsilon 2^{-i}$ and let $A_i' = [a', b)$. We observe that A_i' contains the open interval $A_i'' = (a', b)$ which in turn contains A_i. Letting $A = [a, b)$, by continuity of F there exists $b' < b$ such that $F(b) - F(b') < \varepsilon$. Let $A' = [a, b']$.

Since $\bigcup_{i=1}^{\infty} A_i = A$, the A_i'' cover A and therefore A' as well. Now A' is a closed and bounded interval, and the A_i'' are open sets. By Theorem 20 there exists a finite subcover $\{A_j'' : j \in J\}$. In particular $\bigcup_{j \in J} A_j'$ also covers A'.

We write $\bigcup_{j \in J} A_j' = \bigcup_{j \in J} (A_j' \setminus A_j) \cup A_j$. By the union bound, $L = \bigcup_{j \in J} A_j' \setminus A_j$ has $\mu_0(L) \leq \sum_{i=1}^{\infty} \varepsilon 2^{-i} = \varepsilon$. We now have $L \cup \bigcup_{j \in J} A_j \supseteq A'$, so $\mu_0(L) + \sum_{j \in J} \mu_0(A_j) \geq \mu_0(A')$, and $\mu_0(A') \geq \mu_0(A) - \varepsilon$. Rearranging, we have

$$\sum_{i=1}^{\infty} \mu_0(A_i) \geq \sum_{j \in J} \mu_0(A_j) \geq \mu_0(A) - \varepsilon - \varepsilon\,.$$

Since this holds for all $\varepsilon > 0$ we have $\sum_{i=1}^{\infty} \mu_0(A_i) \geq \mu_0(A)$. So $\sum_{i=1}^{\infty} \mu_0(A_i) = \mu_0(A)$, completing the proof of countable pre-additivity. $\square$

It's easy to believe this theorem should be trivial: 'cancel terms in pairs when half-intervals share an endpoint'. But this isn't a proof when we have infinitely many intervals. Implicitly, doing this we're writing the series in countable pre-additivity as a series with positive and negative terms, which generally is not absolutely convergent. Then to cancel we are rearranging it—and Theorem 18 tells us that can easily change the limit of the series!

This construction produces any *continuous measure* on $[0, 1)$ (with its usual topology!): any measure μ such that $x \to \mu([0, x))$ is a continuous function. Equivalently, there is no x such that $\mu(\{x\}) > 0$.

Exercise 30. *What properties does a function $x \to \mu([0,x))$ have for general (finite, i.e. not taking the value ∞) measures on $[0,1)$? Can you generalise the above theorem to work with any such function, or some subclass of such functions?*

Hint: for any bounded increasing $F : [0,1) \to \mathbb{R}$, prove the left and right limits $\lim_{x\to c-} F(x)$ and $\lim_{x\to c+} F(x)$ exist.

Finally, we can now prove the Lebesgue measure, and other continuous measures, on $\mathbb{R}$, exist.

Let $F : \mathbb{R} \to \mathbb{R}$ be any increasing continuous function. We write $\mathbb{R} = \bigcup_{n\in\mathbb{Z}}[n, n+1)$. For each $n \in \mathbb{Z}$, since $[n, n+1)$ and $[0,1)$ are isomorphic, Theorem 40 returns an algebra $\mathcal{A}_n$ on $[n, n+1)$ and a pre-measure μ_{0n} on $\mathcal{A}_n$ such that $\mu_{0n}([a,b)) = F(b) - F(a)$ for all $n \le a < b \le n+1$. By Theorem 36 this extends to a σ-algebra $\mathcal{F}_n$ and unique measure μ_n on $\mathcal{F}_n$ with the same property. Finally Theorem 38 returns $\mathcal{F}$ on $\mathbb{R}$ and a unique measure μ which agrees with μ_n on each $[n, n+1)$.

Exercise 31. *Prove that we have the property $\mu([a,b)) = F(b) - F(a)$ for all $a < b$, not just when $n \le a < b \le n+1$ for some $n \in \mathbb{Z}$.*

Explain why this construction, applied to the function $F(x) = x$, gives the Lebesgue measure not the Borel measure. Prove that the Borel measure also exists.

Solution to Exercise 31:

Suppose $a < b$ are any two real numbers, and let n be an integer, and m a positive integer, such that $a \le n < b < n+m$. For each $0 \le i \le m-1$, the interval $I_i = [a,b] \cap [n+i, n+1+1) = [s_i, t_i)$ has $\mu(I_i) = F(t_i) - F(s_i)$ by construction (i.e. by Theorem 40, Theorem 36 and Theorem 38). By additivity, the union of the I_i, which is a disjoint union, has measure $\mu([a,b)) = F(b) - F(a)$: this is a finite sum and the interior terms from $t_0, s_1, t_1, \ldots, s_{m-1}$ cancel in pairs.

We can extend this to the case where a or b may be ∞ or $-\infty$ by considering the nested sets $[-n, n] \cap [a,b)$ and applying continuity of measure.

It's actually easiest to first prove Borel measure exists from the above construction. By definition, $\mathcal{B}(\mathbb{R})$ is a subset of $\mathcal{F}$, because $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra containing $\mathcal{A}$. Then $\mu|_{\mathcal{B}(\mathbb{R})}$ is the Borel measure: because it gives all intervals $[a,b)$ their 'correct' measure $b - a$.

In the above construction, when we construct a measure μ' on $\mathcal{F}'$ on $[0,1)$ with Theorem 36, all our countable coverings are countable unions of intervals: so all the countable coverings are in $\mathcal{B}([0,1))$. Given any $F \in \mathcal{F}'$, now we have a measure, copying the proof of Exercise 11(c) we obtain sets A and B in $\mathcal{B}([0,1))$ which cover F and F^c respectively, with respectively $\mu'(A) = \mu'(F)$ and $\mu'(B) = \mu'(F^c)$. So $B^c \subseteq F \subseteq A$, and $F = B^c \cup N$ where N is a subset of $A \setminus B^c$. Since A and B^c have the same measure, N is a μ-null set: and since A and B^c are in $B([0,1))$, N is also Borel-null. So any $F \in \mathcal{F}$ is also in the completion of the Borel measure; the other direction holds by essentially the same logic, and so $\mathcal{F}'$ is the Lebesgue σ-algebra on $[0,1)$; and μ is (by uniqueness) therefore Lebesgue measure. It's easy to check that this goes through Theorem 38.

(The 'point' really here is that we have to generate Borel measure, and we actually put in our σ-algebra all sets whose inner and outer measure are equal: so we have to be getting the measure completion. We don't have a mechanism to get any additional sets, so what we get is exactly the measure completion: Lebesgue measure.)

We can, for example, also now construct the Gaussian measure on $\mathbb{R}$: just take F to be the cumulative distribution function of the Gaussian.